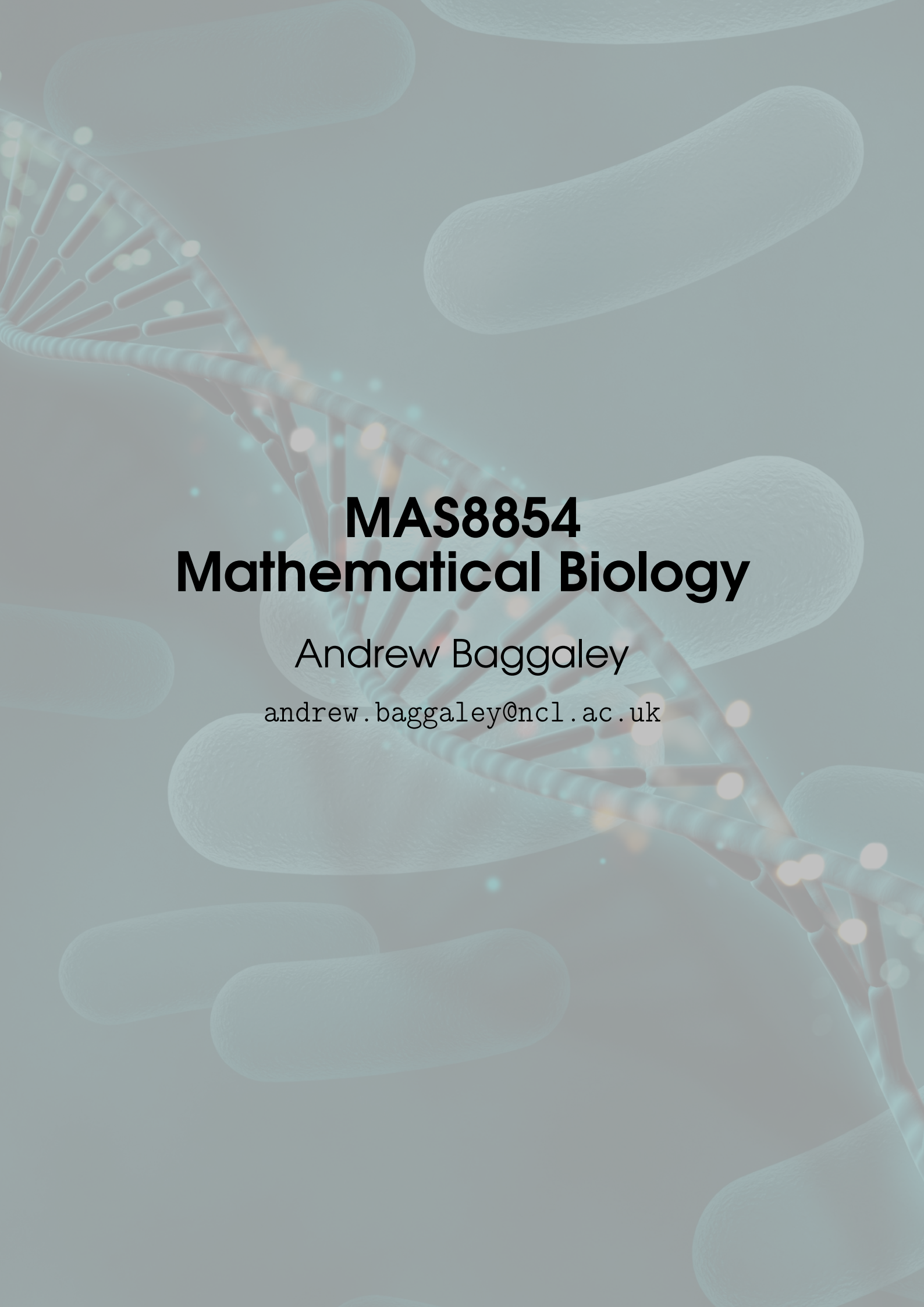




MAS8854

Mathematical Biology

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1. Introduction

1.1 Overview

Mathematical biology is nowadays, a vast field, and the term means many different things to many different people. To get a flavour, please read the following overviews:

- *Mathematics Is Biology's Next Microscope, Only Better; Biology Is Mathematics' Next Physics, Only Better*, J. E. Cohen, PLoS Biol. **2**, e439 (2004).
- *What is mathematical biology and how useful is it?*, A. Friedman, Notices of the American Mathematical Society **5** 851 (2010).

Even if we just restrict ourselves to the interests of the Mathematical Biology Group in Newcastle, the different topics being studied is very diverse, e.g.

- Biological fluid dynamics
- Biomedical ultrasound
- Collective motion in nature
- Microbial population dynamics
- Prehistoric population dynamics
- Stem cells
- Ecological networks
- Tree disease modelling
- Tumour growth

This course will mainly focus on the subject of *mathematical ecology*. This is a personal choice but there are strong reasons for studying this subject:

- it is intrinsically interesting;
- It has important societal ramifications for themes such as climate change, habitat loss, and food security, etc.
- The tools, techniques and models that we will learn in this context are broadly applicable to many other problems in mathematical biology (and beyond).

1.2 Literature

The majority of this course is based around two textbooks:

- *Mathematical Biology I*, J. D. Murray (Springer 2002). This is the classic textbook in the field of mathematical biology. There is a part II as well.

- *Elements of Mathematical Ecology*, M. Kot (Cambridge 2001). This great book gives a pedagogical introduction to mathematics in ecology and of population dynamics.

Other useful books on theoretical/mathematical ecology include

- *Theoretical ecology: principles and applications*, R. May and A. R. Mclean (Eds.) (OUP 2007). Seminal book; great overview of many important topics in theoretical ecology.
- *Community Ecology*, G. G. Mittelbach (Oxford 2012). Provides more of an ecologists perspective on this material

1.3 Course structure

This course will have a slightly looser structure, in comparison to previous modules. We will not make a distinction between lectures and problem classes, but will tackle end of chapter questions as and when we meet them. Lectures are:

- Monday 3-4, TR1
- Tuesday 11-12, TR2
- Thursday 9-10, TR2

Office hours are Thursday 10-11 (in 2.09A), feel free to call by whenever, or email to make an appointment.

There are two pieces of assessed coursework the deadlines for these are:

- Assessment 1, **Thursday 21st February, 4pm**
- Assessment 2, **Thursday 7th March, 4pm**

Assessment material will be available through the course blackboard page, however the best place to go for all material is my personal website:

Exponential growth
Logistic growth
General stability considerations
No oscillations
Dimensionless variables
Delay models

Delay can give population oscillations
Linear Stability of DDEs

2. Population dynamics: continuous

Wikipedia:

Population dynamics is the branch of life sciences that studies the size and age composition of populations as dynamical systems, and the biological and environmental processes driving them (such as birth and death rates, and by immigration and emigration). Example scenarios are ageing populations, population growth, or population decline.

We begin this course by considering the dynamics of a single population (without age structure), and concern ourselves with the growth, decay and stability of this population. Our initial approach is to look at this problem as one continuous in time. The mathematical apparatus we will employ is the ordinary differential equation (ODE) and its time-delayed counterpart, the delay differential equation (DDE).

2.1 Exponential growth

We begin by considering an homogeneous population of $N(t)$ individuals. We assume that all changes arise from a per-capita birth rate b and a per-capita death rate d , both of which are assumed constant. The rate of change of population number is then

$$\frac{dN}{dt} = (b - d) N = rN, \quad (2.1)$$

where $r \equiv b - d$ is the *intrinsic rate of growth* of the population. This can be rewritten as

$$\frac{1}{N} \frac{dN}{dt} = r, \quad (2.2)$$

which illustrates that the per-capita growth rate is constant. Eq. (2.1) is an ordinary differential equation (ODE), and its solution is

$$N(t) = N_0 e^{rt}, \quad (2.3)$$

with $N_0 = N(t = 0)$, the initial population. Thus, the population either grows or shrinks exponentially. The exponential growth of populations with $r > 0$ is often referred to *Malthusian*

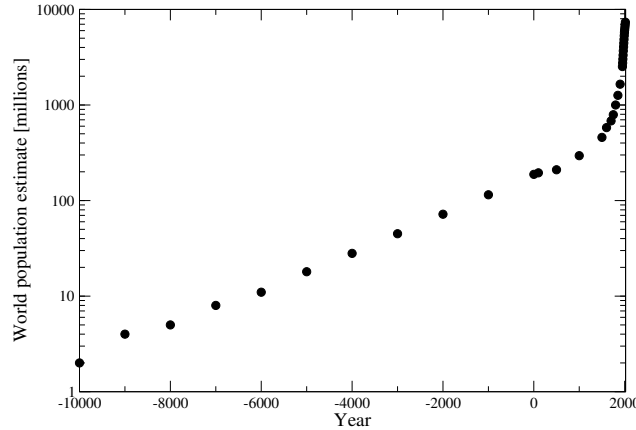


Figure 2.1: Estimated world population based on data from wikipedia. For much of history, growth appears exponential with a growth rate of $r \approx 1.03$. In recent times, the rate of growth rate has increased dramatically [this can actually be fit quite well with piecewise exponential growth with different exponents mapping onto different historical eras, see: [this blog post](#)].

growth after Thomas Robert Malthus who in the 18th century wrote one of the earliest and most influential books on population¹. Fig. 2.1 plots the world population in light on this model.

Inserting the value $N(t) = 0$ into Eq. (2.1) gives $dN/dt = 0$ and the population will not move away from this value. This is a simple example of a *fixed point* (a.k.a. equilibrium point, critical point). We will typically use a star to indicate a fixed point. For this model, the fixed point $N_\star = 0$ is *unstable*, which means that if the population is disturbed even slightly away from this value ($N(t) \geq 0$) the population will move away from this starting value.

2.2 Logistic growth

The above model with its constant birth and death rates is unrealistic as it predicts limitless growth. A perhaps more realistic model is afforded by the *logistic equation*²

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) \quad (2.4)$$

in which K is a constant called the *carrying capacity*. The idea is that members of the population need resources and that increased demand for resources leads to a decrease in the growth rate. Indeed, the per-capita growth rate

$$\frac{1}{N} \frac{dN}{dt} = r \left(1 - \frac{N}{K} \right) \quad (2.5)$$

decreases with increasing population until it reaches zero when $N = K$. This is an example of density-dependent (per-capita) growth rate, since the right-hand side of Eq. (2.5) depends on the density.

¹The mismatch between geometric (here, exponential) population growth and only arithmetic growth of resources (in particular food) is the essential driver behind disease, starvation, war, resulting in what is called a Malthusian catastrophe.

²Also referred to as the *Verhulst equation* after Pierre Verhulst who studied it in this context, 1844-45

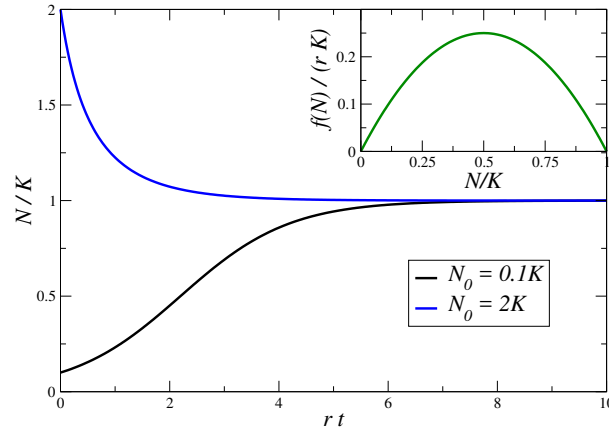


Figure 2.2: MAIN PANEL: The population $N(t)$ according to the logistic equation for two different initial conditions $N_0/K = 0.2, 2$. For all $N_0 > 0$ the population approaches the equilibrium point $N = K$. INSET: The function $f(N)$ that defines the logistic equation as a function of N/K .

The logistic equation has two fixed points: $N_{\star 1} = 0$ and $N_{\star 2} = K$. This can be seen from the inset of Fig. 2.2 which shows the righthand side of Eq. (2.4). What about the stability of these fixed points? Sufficiently close to $N_{\star 1}$, we must have $N^2/K \ll N$ so that

$$\frac{dN}{dt} \approx rN \quad (2.6)$$

For $r > 0$, small perturbation makes population grow exponentially away from $N_{\star} = 0$ as before. This fixed point is therefore still unstable.

Close to $N_{\star 2}$, let us define $n = N - K$ as the deviation of the population away from the fixed-point value. Placing this into Eq. (2.4) gives

$$\frac{dn}{dt} = -rn - \frac{r}{K}n^2 \approx -rn \quad (2.7)$$

where we have approximated the right-hand side with the linear term only since, close to K , the quantity n will be small. The solution to this is

$$n(t) \approx n_0 e^{-rt}, \quad (2.8)$$

and, with $r > 0$, small perturbations about $N_{\star 2}$ will *decay* exponentially. Thus $N_{\star 2} = K$ is a *stable* fixed point.

Fig. 2.2 shows the time evolution of two initial populations governed by the logistic equation with $r > 0$. In the long time limit $rt \rightarrow \infty$ both populations tend towards the fixed point $N_{\star 2}$. Unless we start with $N = 0$, this is always the case in this model.

Exercise 2.1 Find an exact solution for the logistic equation of Eq. (2.4) with starting population N_0 at time $t = 0$. ■

2.3 General stability considerations

Consider a single-species population described by the ODE model

$$\frac{dN}{dt} = f(N). \quad (2.9)$$

Definition 2.3.1 N_* is a fixed point of Eq. (2.9) if $f(N_*) = 0$

Definition 2.3.2 Fixed point N_* is called *feasible* if $N_* \geq 0$.

Clearly, only feasible fixed points correspond to meaningful biological populations.

Definition 2.3.3 Fixed point N_* is *Lyapunov stable* if for any $\epsilon > 0$ there exists a $\delta > 0$ (which depends on ϵ) such that, for all initial conditions $N(t_0) = N_0$ satisfying $|N_0 - N_*| < \delta$, we have $|N(t) - N_*| < \epsilon$ for all $t > t_0$.

Roughly speaking, a fixed point is Lyapunov stable if starting close to that point guarantees that we remain close to it at all later times.

Definition 2.3.4 Fixed point N_* is *asymptotically stable* if it is Lyapunov stable and there exists a $\rho > 0$ such that

$$\lim_{t \rightarrow \infty} |N(t) - N_*| = 0, \quad (2.10)$$

for all N_0 satisfying

$$|N_0 - N_*| < \rho. \quad (2.11)$$

If a system is asymptotically stable, then, small perturbations of the system away from the fixed point cause the system to eventually return to that point.

Theorem 2.3.1 Consider a fixed point N_* of Eq. (2.9) with continuously-differentiable $f(N)$. Fixed point N_* is (asymptotically) stable if $f'(N_*) < 0$ and unstable if $f'(N_*) > 0$.

Proof. (see Kot, p10) Define $n = N(t) - N_*$. Expand $f(N)$ about N_* we obtain

$$\frac{dn}{dt} = f'(N_*)n + g(n) \quad (2.12)$$

where $g(n)$ contains all terms above linear and satisfies $g(0) = 0$ and $g'(0) = 0$. This (plus continuity of g' , from continuity of f) guarantees that for each $\epsilon > 0$ there is a small δ neighbourhood about zero where $|g'(n)| < \epsilon$. Thus,

$$g(n) = \int_0^n g'(s)ds \leq \epsilon|n|. \quad (2.13)$$

It follows that

$$\frac{dn}{dt} \leq f'(N_*)n + \epsilon|n|. \quad (2.14)$$

For small enough δ and ϵ and $f'(N_*) \neq 0$, the higher-order terms cannot change the sign of dx/dt . As a result a small enough perturbation will decay and the fixed point is asymptotically stable. A similar statement can be made to show that $f'(N_*) > 0$ implies instability. ■

Example 2.1 Consider the population model

$$\frac{dN}{dt} = rN + m, \quad (2.15)$$

where r is the growth rate and m a rate of migration. Find fixed points of this model and the parameters for which the stationary populations are *feasible*. Investigate the stability of these points.

Solution. Stationarity, $dN/dt = 0$, delivers immediately that

$$N_* = -m/r, \quad (2.16)$$

and this is the only fixed point of the model. For N_* to be feasible clearly requires m and r have opposite signs. The stability of the fixed point determined by $f'(N_*) = r$ and thus the sign of r determines the stability: $r < 0$ implies stability and $r > 0$ instability. Thus the only parameter set which delivers a feasible, stable (nontrivial) stationary state is when $r < 0$ and $m > 0$, such that immigration compensates for population decline. ■

Exercise 2.2 The Gompertz equation (Gompertz, 1825)

$$\frac{dN}{dt} = \alpha N \log\left(\frac{K}{N}\right) \quad (2.17)$$

is an alternative growth model used, amongst other things, in modelling the growth of tumours. Find the fixed points of this model and their stability ($\alpha > 0$). ■

2.4 No oscillations

One important statement that we can make about the class of models defined by Eq. (2.9) is they can not exhibit periodic behaviour.

Definition 2.4.1 A function $N: \mathbb{R} \rightarrow \mathbb{R}$ is *periodic* with period T if:

- $N(t + T) = N(t) \forall t \in \mathbb{R}$;
- T is the smallest value for which this condition holds and $T > 0$

The second condition means that constant functions are not regarded as periodic.

Theorem 2.4.1 If $N(t)$ satisfies the ODE Eq. (2.9) for some continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$, then $N(t)$ cannot be periodic.

Proof. Proof by contradiction (Murray, p17). Assume that $N(t)$ is periodic. Multiply each side of Eq. (2.9) with dN/dt and integrate over a single period

$$\int_0^T \left(\frac{dN}{dt}\right)^2 dt = \int_0^T f(N) \frac{dN}{dt} dt. \quad (2.18)$$

Since $N(t)$ is non-constant, dN/dt must be non-zero somewhere in the interval $[0, T]$. Thus $(dN/dt)^2$ is strictly positive somewhere in this same interval and the complete integral on the lefthand side is also strictly positive. In contrast, we can rewrite the righthand side as

$$\int_0^T f(N) \frac{dN}{dt} dt = \int_{N(0)}^{N(T)} f(N) dN = \int_{N(0)}^{N(0)} f(N) dN = 0, \quad (2.19)$$

where we have used the (assumed) periodic nature of $N(t)$ to rewrite the limits. Thus, we obtain a contradiction and therefore N can not be periodic. ■

2.5 Dimensionless variables

It is often useful to introduce dimensionless variables and otherwise rescale populations, as this often simplifies a problem. Consider the logistic equation Eq. (2.4). Let us define

$$x = \frac{N}{K} \quad \text{and} \quad \tau = rt. \quad (2.20)$$

The first is the population expressed as a fraction of the carrying capacity; the second is a *dimensionless time* variable obtained by multiplying the growth rate (dimensions: 1/time) by the time elapsed (dimensions: time). Using the chain rule (twice), we have

$$\frac{dN}{dt} = \frac{dN}{dx} \frac{dx}{d\tau} \frac{d\tau}{dt} = Kr \frac{dx}{d\tau}. \quad (2.21)$$

This allows us to rewrite Eq. (2.4) as

$$\frac{dx}{d\tau} = x(1 - x). \quad (2.22)$$

This form is simpler. Moreover, it has no parameters at all. This means that we do not need to analyse the equation for various values of K and r — there is just a single underlying system to worry about, namely Eq. (2.22).

2.6 Delay models

One shortcoming of the models in the previous sections is that they do not take into account phenomena such as gestation or the time taken to reach sexual maturity. To do so necessitates a move away from ODEs and on to the study of *delay-differential equations* (DDEs). Perhaps the simplest generalisation is to the class of models

$$\frac{dN}{dt} = f[N(t), N(t - T)], \quad (2.23)$$

where $T > 0$ is some fixed delay time. For example, we can modify the logistic equation to give

$$\frac{dN}{dt} = rN(t) \left[1 - \frac{N(t - T)}{K} \right], \quad (2.24)$$

which is known as the *Hutchinson-Wright model* (or delayed logistic equation). Biologically we might justify this model as follows. For egg-laying species, clutch size is presumably determined not by the concentration of unconsumed food available when the eggs hatch, but by the amount of food available when the eggs were forming. Thus, the effects of resource-competition will be delayed some time T , resulting in the above modification of the carrying-capacity term as above.

A more general class of DDE is provided by a Volterra integro-differential equation

$$\frac{dN}{dt} = rN(t) \int_0^\infty \kappa(t') \left[1 - \frac{N(t - t')}{K} \right] dt'. \quad (2.25)$$

Here the *memory kernel* $\kappa(t)$ weights the different historical contributions and obeys

$$\int_0^\infty \kappa(t) dt = 1. \quad (2.26)$$

A δ -peaked kernel $\kappa(t) = \delta(t - T)$ recovers the Hutchinson-Wright model

$$\frac{dN}{dt} = rN(t) \int_0^\infty \delta(t' - T) \left[1 - \frac{N(t - t')}{K} \right] dt' = rN(t) \left[1 - \frac{N(t - T)}{K} \right]. \quad (2.27)$$

2.6.1 Delay can give population oscillations

It should not be underestimated what a drastic change the move to DDEs is! For ODEs, a particular solution is specified by the ODE itself plus the value of the population $N(0)$ at a particular time $t = 0$. For a DDE like Eq. (2.23), the situation is much more complicated. To specify behaviour on interval $0 < t < T$, we need to know initial data for the whole interval $-T < t < 0$ as the system has a “memory” of what happened in the past.

One new feature that arises with DDEs is the possibility of oscillations, as the following example demonstrates.

Consider a model like Eq. (2.15) but where the animals have gestation period T , so that the number of animals being born at time t depends on the population at time $t - T$, whilst the death term is still time-local. We have

$$\frac{dN}{dt} = bN(t - T) - dN(t) - m, \quad (2.28)$$

with per-animal birth rate b , death rate d and outward migration rate $m > 0$.

We begin by rescaling with

$$x = \frac{N}{N_\star} = \frac{N(b-d)}{m} \quad \text{and} \quad \tau = bt \quad (2.29)$$

to give

$$\frac{dx}{d\tau} = x(t - \tilde{T}) - \tilde{d}x(\tau) - (1 - \tilde{d}) \quad (2.30)$$

where $\tilde{T} = bT$ is the dimensionless delay time and $\tilde{d} = d/b$. We will assume $b > d > 0$, so that $0 < \tilde{d} < 1$.

Since we are looking for oscillations, let's try the Ansatz

$$x(\tau) = A + B \cos(\omega\tau) \quad (2.31)$$

Substitution into DDE Eq. (2.30) gives

$$\begin{aligned} -\omega B \sin(\omega\tau) &= (1 - \tilde{d})A - (1 - \tilde{d}) + B \cos(\omega\tau) [\cos(\omega\tilde{T}) - \tilde{d}] \\ &\quad + B \sin(\omega\tau) \sin(\omega\tilde{T}). \end{aligned} \quad (2.32)$$

Equating coefficients on the left and right gives

$$(1 - \tilde{d})A - (1 - \tilde{d}) = 0; \quad (2.33)$$

$$\cos(\omega\tilde{T}) - \tilde{d} = 0; \quad (2.34)$$

$$-\omega = \sin(\omega\tilde{T}). \quad (2.35)$$

These equations are only satisfied for certain values of A , ω , $\tilde{T}$ and $\tilde{d}$. Since $0 < \tilde{d} < 1$, Eq. (2.34) means $\cos(\omega\tilde{T}) > 0$. Similarly, since $\omega > 0$, from Eq. (2.35) we have $\sin(\omega\tilde{T}) < 0$. These constraints are satisfied when $3\pi/2 < \omega\tilde{T} < 2\pi$. Choosing a value for $\omega\tilde{T}$ then determines $\tilde{d}$ via Eq. (2.34) and ω via Eq. (2.35). From Eq. (2.33) we have $A = 1$, and amplitude B may be freely chosen.

This example is somewhat contrived. Nevertheless it demonstrates that, in contrast to the ODE single-species population models, DDE models can exhibit oscillations.

2.6.2 Linear Stability of DDEs

The method of linearisation about the fixed points is also applicable to DDEs. We discuss this through the example of the Hutchinson-Wright equation.

We start with

$$\frac{dN}{dt} = rN(t) \left[1 - \frac{N(t-\tau)}{K} \right] \quad (2.36)$$

and rescale with

$$x = N/K \quad \text{and} \quad \tau = rt \quad (2.37)$$

to give

$$\frac{dx}{d\tau} = x(\tau) [1 - x(\tau - \tilde{T})] \quad (2.38)$$

with $\tilde{T} = rT$. To find the fixed points we note that $\frac{dx}{d\tau} = 0$ implies that $x(\tau + \tilde{T}) = x(\tau)$. Thus, the fixed points here are the same as for the undelayed logistic equation, namely $x(\tau) = 0$ and $x(\tau) = x_\star = 1$. We investigate stability of the second solutions by writing

$$x(\tau) = x_\star + \eta(\tau) = 1 + \eta(\tau), \quad (2.39)$$

where $\eta(\tau)$ is some “suitably small” function. Substituting this into the DDE we have

$$\frac{d\eta}{d\tau} = [1 + \eta(\tau)] [1 - \{1 + \eta(\tau - \tilde{T})\}]. \quad (2.40)$$

We then linearise by keeping lowest order term in η

$$\frac{d\eta}{d\tau} \approx -\eta(\tau - \tilde{T}). \quad (2.41)$$

This looks similar to a linear first-order ODE, so we try an exponential Ansatz

$$\eta(t) = \eta_0 e^{\lambda \tau}, \quad (2.42)$$

which results in the equation

$$\lambda = -e^{-\lambda \tilde{T}}. \quad (2.43)$$

The graphical solution of this equation is illustrated in Fig. 2.3, and depending on the value of $\tilde{T}$, we obtain either 0, 1 or 2 real solutions for λ . The repeated root is of interest as this marks the boundary between real and complex λ . The repeated root occurs when the two curves $f(\lambda) = \lambda$ and $f(\lambda) = -e^{-\lambda \tilde{T}}$ are tangential to one another. Equating the gradients, we have

$$1 = \tilde{T} e^{-\lambda \tilde{T}}. \quad (2.44)$$

Taken together Eq. (2.43) and Eq. (2.44) mean that

$$\lambda \tilde{T} = -1; \quad \tilde{T} = 1/e \quad (2.45)$$

Below $\tilde{T} = 1/e$, then, Eq. (2.43) has two real roots.

Above this value, we expect λ to be complex, so let us write $\lambda = \mu \pm i\omega$. These real and imaginary components are plotted in Fig. 2.3. It is clear that μ crosses through zero and at the crossing, λ is purely imaginary. Let us investigate this analytically. With $\lambda = i\omega$, the real part of Eq. (2.43) yields $\cos(\omega \tilde{T}) = 0$ and therefore $\omega \tilde{T} = (2n + 1)\pi/2$. From the numerics, we see that it is the lowest, $n = 0$ solution that we seek, and therefore $\omega \tilde{T} = \pi/2$. Substituting this into the imaginary part of Eq. (2.43) (with $\lambda = i\omega$) gives $\omega = 1$. Thus, we have shown that a *oscillatory solution* exists when $\tilde{T} = \pi/2$.

Increase of $\tilde{T} > \pi/2$ gives solutions for which both $\mu > 0$ and $\omega > 0$. This means that there exists oscillatory *instabilities* that grow exponentially. To investigate the full dynamics, we must then resort to numerical solution of the DDE. Fig. 2.4 shows some typical traces for illustrative values of $\tilde{T}$. For $\tilde{T} > \pi/2$, the behaviour settles down at long times to a periodic behaviour.

Conclusion: delay into single population models can cause instabilities and oscillations.

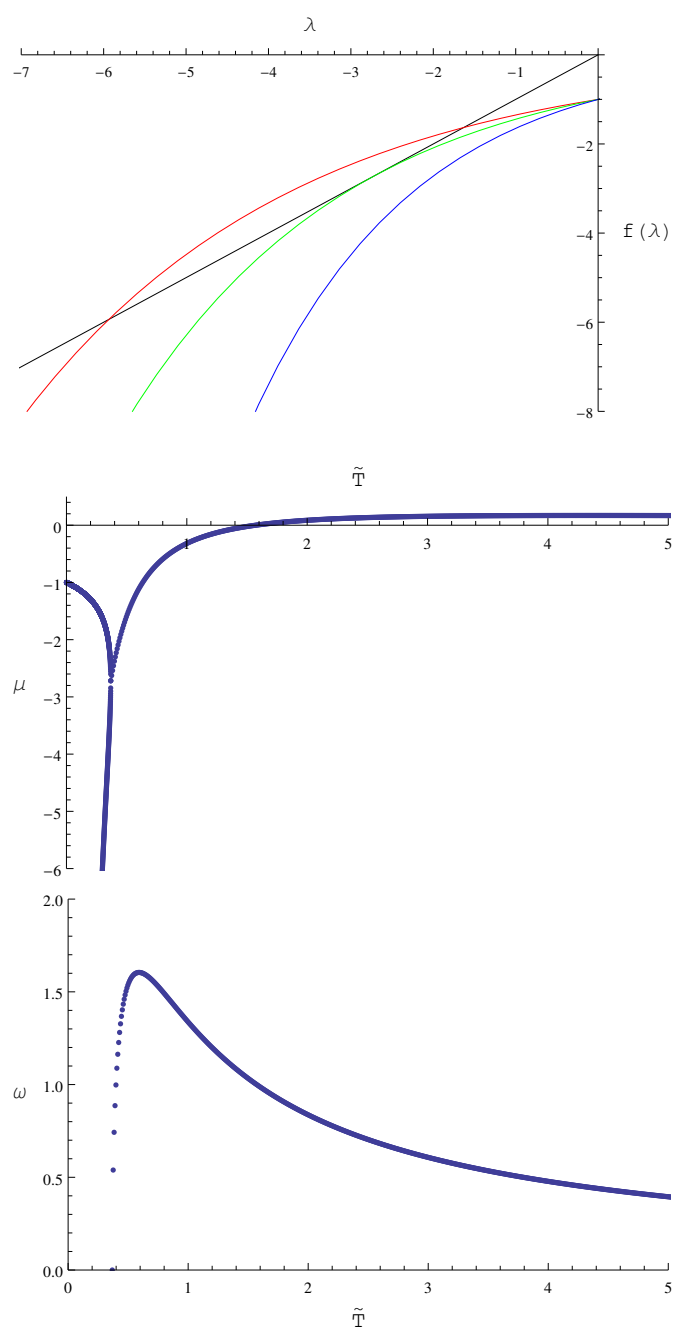


Figure 2.3: TOP: Graphical solution of Eq. (2.43). The black line is $f(\lambda) = \lambda$ and the coloured lines are $f(\lambda) = -e^{-\lambda\tilde{T}}$ for $\tilde{T} = 0.3$ (red), $\tilde{T} = 1/e = 0.367 \dots$ (green), $\tilde{T} = 0.5$ (blue). Solutions exist where the black line crosses when appropriate coloured one. MIDDLE: Real part μ of λ as a function of $\tilde{T}$. BOTTOM: Imaginary part ω of λ as a function of $\tilde{T}$.

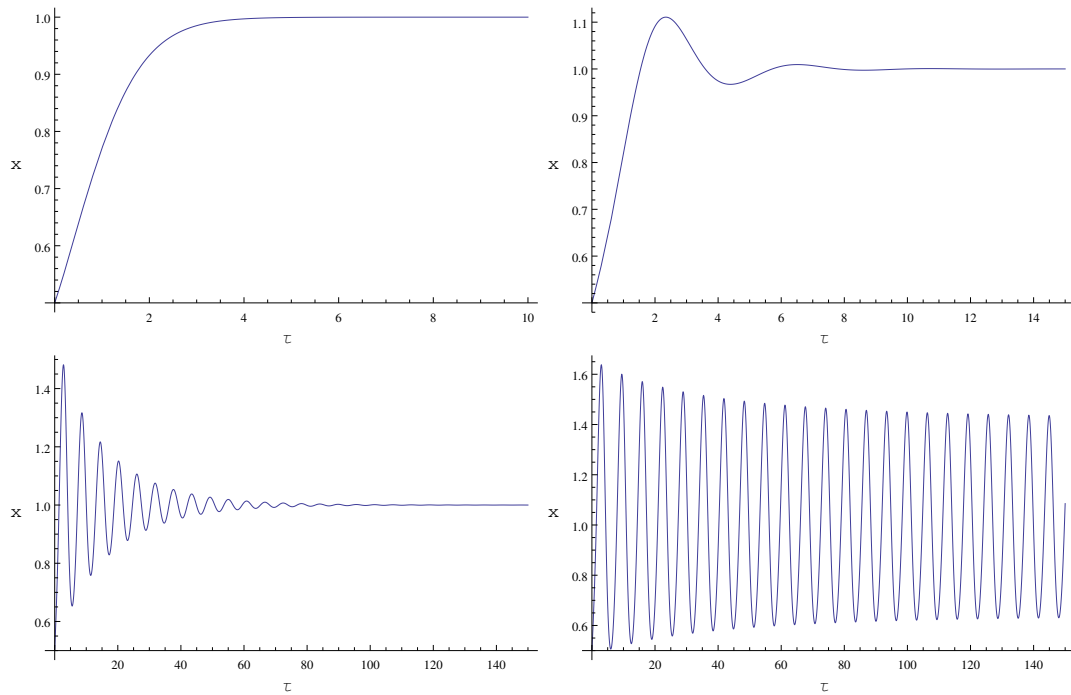


Figure 2.4: The population $x(\tau)$ in the (scaled) Hutchinson-Wright model for delay parameters: $\tilde{T} = 0.3$ (top left), $\tilde{T} = 0.8$ (top right), $\tilde{T} = 1.4$ (bottom left), $\tilde{T} = 1.6$ (bottom right). In the first instance λ is real and the solution damps. In the second two cases, λ is complex with a negative real part, and correspondingly, damped oscillations are observed. In the final case, with $T > \pi/2$, λ is complex with positive real part. The fixed point thus becomes unstable, and as the numerics show, a periodic limit cycle develops. Here initial conditions of $x(\tau) = 0.5$ for $t \leq 0$ were assumed.

3. Population dynamics: discrete

Differential equation models imply a continuous overlap of generations. Many species have no overlap whatsoever between successive generations and so population growth is in discrete steps. Depending on the time between generations and the time-scale over which we are interested, a continuous-time model may be a reasonable approximation. In other circumstances, however, this discreteness in generation is important. In this case, we are led to the study of discrete-time models, and in particular difference equations.

3.1 Geometric growth and decay

Let N_t be the size of population in year (or generation) t . Each individual leaves R_0 offspring before dying. Then

$$N_{t+1} = R_0 N_t \quad (3.1)$$

with R_0 the *net reproductive rate*. Eq. (3.1) is a first-order linear difference equation. The solution is simply

$$N_t = R_0^t N_0, \quad (3.2)$$

i.e. one of geometric growth or decay depending on whether R_0 is greater or smaller than one. This result is similar to that found for the continuous model of Ch. 2. Indeed, if we identify $R_0 = e^r$ then Eq. (3.2) reads $N_t = e^{rt} N_0$, which may be directly compared with Eq. (2.3).

The fixed points of difference equation clearly obey $N_{t+1} = N_t = N_*$, $\forall t$. For model Eq. (3.1) this means $0 = (1 - R_0)N_t$. Thus, unless $R_0 = 1$ (in which case all values of N are fixed), the only fixed point is the origin $N_* = 0$.

To see whether this point is stable or not we introduce in Fig. 3.1 a graphical technique for tracking solutions of a difference equation called “cobwebbing”.

Exercise 3.1 Perform a cobweb analysis of Eq. (3.1) with $R_0 = 1/2$

As this cobweb analysis should make clear, under Eq. (3.1), the origin is stable for $0 < R_0 < 1$ and unstable for $R_0 > 1$.

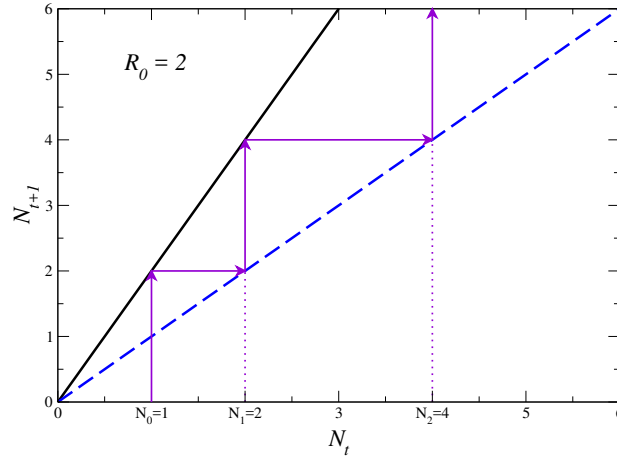


Figure 3.1: Analysis of Eq. (3.1) with the cobwebbing technique. The black line is Eq. (3.1) with $R_0 = 2$; the blue dashed line is $N_{t+1} = N_t$. We iterate the difference equation by moving vertically to the black curve, and from that point horizontally to the blue 45deg line. These steps are shown in purple. In this case we have geometric growth.

3.2 Stability of difference equations

The notions of Lyapunov and asymptotic stability are similar here to the continuous-time case. The criteria for stability is, however, somewhat modified:

Theorem 3.2.1 Consider the first-order autonomous difference equation

$$N_{t+1} = F(N_t), \quad (3.3)$$

with continuously-differentiable function $F(N)$. A fixed point N_* of Eq. (3.3) is asymptotically stable if $|F'(N_*)| < 1$ and unstable if $|F'(N_*)| > 1$.

Proof. As continuous case, except that the final linearised equation is a difference equation with corresponding criteria for growth or decay as in Sec. 3.1. ■

Example 3.1 Find the fixed point of the difference equation

$$N_{t+1} = aN_t + b \quad (3.4)$$

and describe its feasibility and stability in terms of real parameters a and b .

Solution. For $a \neq 1$, setting $N_{t+1} = N_t = N_*$ gives $N_* = b/(1 - a)$. Also $F(N) = aN + b$ so that $F'(N) = a$. Thus, feasibility of the stationary population requires either: $b > 0$ and $a < 1$, in which case the fixed point is stable when $|F'| = |a| < 1$; or $b < 0$ and $a > 1$ in which case the fixed point is unstable ($|F'| = |a| > 1$). For $a = 1$ no fixed point exists — this can be confirmed by a cobweb plot where we see that the $F(N)$ never intercepts the $N_{t+1} = N_t$ line. ■

Exercise 3.2 The Beverton-Holt model was introduced to describe fishery populations. It reads

$$N_{t+1} = \frac{R_0 N_t}{1 + \alpha N_t}; \quad \alpha = (R_0 - 1)/K \quad (3.5)$$

with $R_0 > 0$ a net growth rate per generation and $K > 0$ a carrying capacity. Find the fixed points of this model and determine their stability. ■

3.3 The Logistic Map and Chaos

The logistic map is as a discrete-time demographic model analogous to the logistic equation. Robert May's seminal 1976 paper used this model to highlight how complex, chaotic behaviour can arise from very simple non-linear dynamical equations.

The logistic map can be derived from the logistic equation. We start with Eq. (2.4) and approximate the differential with a forward-difference quotient

$$\frac{N_{t+\Delta t} - N_t}{\Delta t} = rN_t \left(1 - \frac{N_t}{K}\right) \quad (3.6)$$

Scaling the time so that for one generation $\Delta t = 1$, we obtain

$$N_{t+1} = (1 + r)N_t - \frac{r}{K}N_t^2. \quad (3.7)$$

We can then tidy things up by rescaling $x_t \equiv r/(1 + r)N_t/K$ and $\mu = 1 + r$ to give the *logistic map*

$$x_{t+1} = \mu x_t(1 - x_t). \quad (3.8)$$

This map has a single parameter μ , the interesting range for which is $\mu \in [0, 4]$. This simple map displays a surprisingly rich suite of behaviours.

Exercise 3.3 Show that the fixed points of the logistic map are

$$x_\star = 0 \quad \text{and} \quad x_\star = (\mu - 1)/\mu \quad (3.9)$$

and show that the stability of $x_\star = 0$ changes from stable to unstable at $\mu = 1$ and that $x_\star = (\mu - 1)/\mu$ is only stable for $1 < \mu < 3$. ■

Based on this analysis we conclude that with μ between 0 and 1, the population will eventually die, independently of the initial population, since the origin is the only stable fixed point. For μ between 1 and 3, however, the population will approach the finite value $(\mu - 1)/\mu$, again independently of the initial population (with damped oscillations for $2 < \mu < 3$ due to the negative sign of $F'(x)$). These behaviours are observed in Fig. 3.2.

3.3.1 Period doubling

Above $\mu = 3$, however, both the above fixed points becomes unstable. What happens in this regime? To answer this, we first look at the numerics. The uppermost plot in Fig. 3.3 shows the behaviour for $\mu \approx 3.44949$ and we observe that in the long-time limit, the population approaches a permanent oscillation between two distinct values (the values of which depend on μ). This *period-doubled* behaviour persists throughout the range $3 < \mu < 1 + \sqrt{6}$.

This behaviour can be understood by considering the doubly-iterated map $x_{t+2} = F^2(x_t) = F(F(x_t))$ and looking for its fixed points.

Exercise 3.4 With $F(x) = \mu x(1 - x)$, show that

$$\begin{aligned} -x + F^2(x) &= -x + F(F(x)) \\ &= -\mu x \left(x - \frac{\mu - 1}{\mu} \right) [\mu^2 x^2 - \mu(1 + \mu)x + 1 + \mu] \end{aligned} \quad (3.10)$$

Hence show that the fixed points of the double-iterated logistic map

$$x_{t+2} = F^2(x_t) \quad (3.11)$$

are

$$x_{\star 0} = 0; \quad x_{\star 1} = (\mu - 1)/\mu; \quad x_{\star \pm} = \frac{1}{2\mu} \left[1 + \mu \pm \sqrt{(\mu - 3)(\mu + 1)} \right]. \quad (3.12)$$

The first two fixed points are clearly those of the original map F . The fixed points $x_{\star \pm}$ are new and it is clear that they are only real for $\mu > 3$. We can then look at the stability of these points

$$(F^2(x_{\star \pm}))' = F'(F(x_{\star \pm}))F'(x_{\star \pm}) = 4 + 2\mu - \mu^2. \quad (3.13)$$

The stability criterion is then

$$|4 + 2\mu - \mu^2| < 1 \quad \Rightarrow \quad \mu < 1 + \sqrt{6}. \quad (3.14)$$

Thus, the 2-cycle is extant and attractive in the range $3 < \mu < 1 + \sqrt{6}$.

3.3.2 Route to chaos

Increasing μ further we see more bifurcations. Period-4 behaviour is observable in the range $3.44949 < \mu < 3.54409$. We can get this parameter range from consideration of the four-times-iterated map $F^4(x_t)$, but finding the fixed points from this requires finding the roots of a 12th-degree polynomial. Beyond this, period-8, then period-16, and so forth is observed. This behaviour is an example of a period-doubling cascade. At a value of $\mu = 3.56995$ and for most values above it, we have the onset of dynamical chaos — for almost all initial conditions, we no longer see oscillations of finite period; slight variations in the initial population yield dramatically different results over time. This is the so-called “butterfly effect”, a prime characteristic of chaos. This period-doubling sequence and onset of chaos is often visualized with a *bifurcation diagram* as in Fig. 3.4. This diagram shows the points of the attractor at each value of μ .

What is particularly interesting is that this period-doubling route to chaos is not a unique feature of the logistic map. Rather it can be shown that any uni-modal map (i.e. with a single “hump”) will show such behaviour and that the sequence of bifurcations exhibits certain universal feature. See Feigenbaum constants.

At this point, you should read the article

- *Simple mathematical models with very complicated dynamics*, R. M. May, Nature **261** 459 (1976).

3.4 Delayed difference equations

The difference in behaviour between the logistic equation and the logistic map is marked. In part this difference arises from the implicit delay in discrete case: behaviour at time $t + 1$ depends on behaviour at time t . Moreover, we have already seen that adding delays in the continuous case can increase the richness of behaviour. What happens when we include explicit delays into discrete population models?

An obvious model with which to investigate such questions is the discrete-time version of the Hutchinson-Wright model Eq. (2.24)

$$N_{t+1} = N_t \left[1 + r \left(1 - \frac{N_{t-T}}{K} \right) \right], \quad (3.15)$$

which we might call the *delayed logistic map*. Analysis of this model forms part of the module assignment. Another famous example is the *delayed Beverton-Holt* model

$$N_{t+1} = \frac{R_0 N_t}{1 + [(R_0 - 1)/K] N_{t-T}}. \quad (3.16)$$

Suffice to say, the behaviour of these delayed difference equations is complicated!

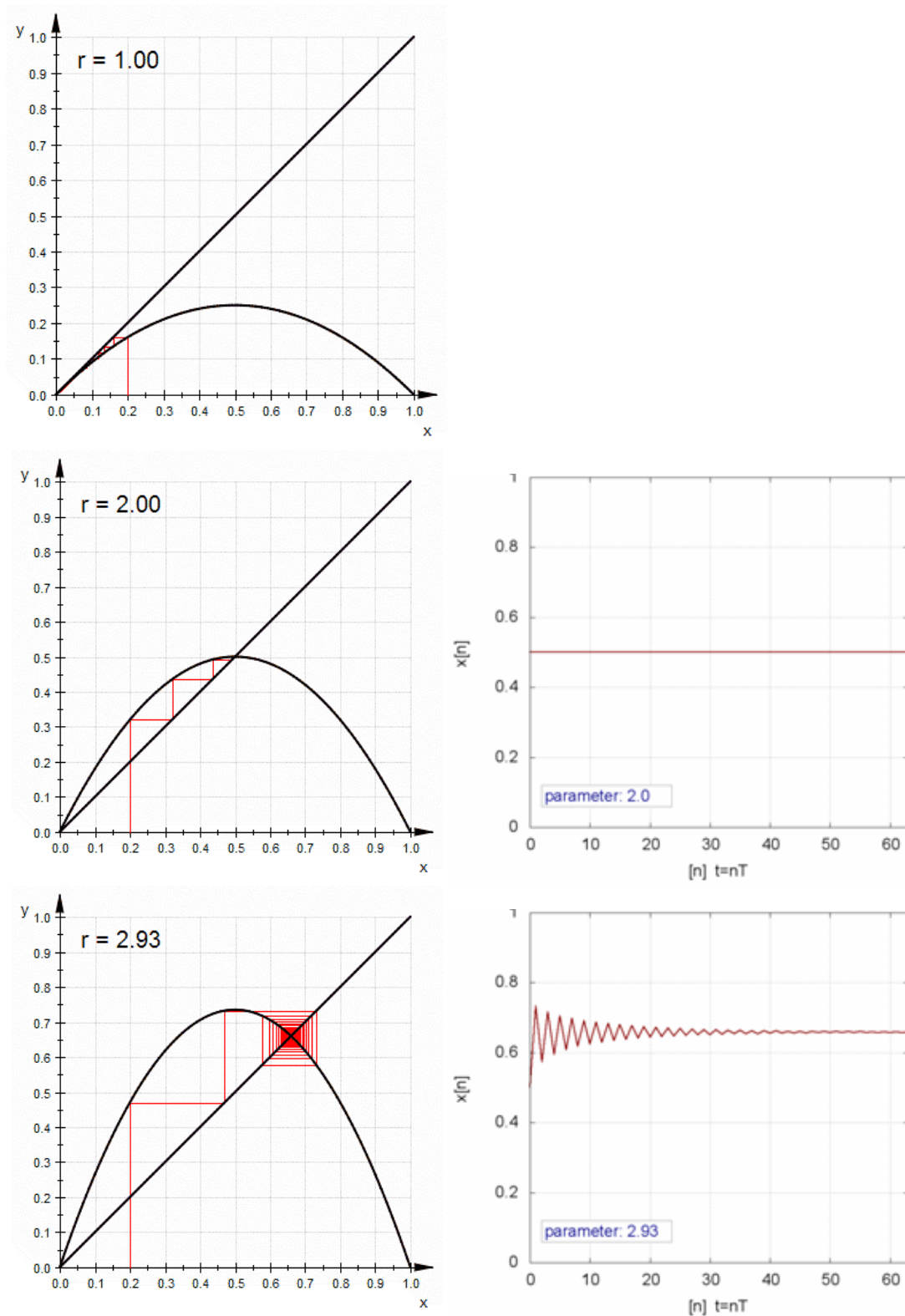


Figure 3.2: Behaviour of the logistic map for parameter values $\mu = 1$, $\mu = 2$, and $\mu = 2.93$. Figures from wikipedia.

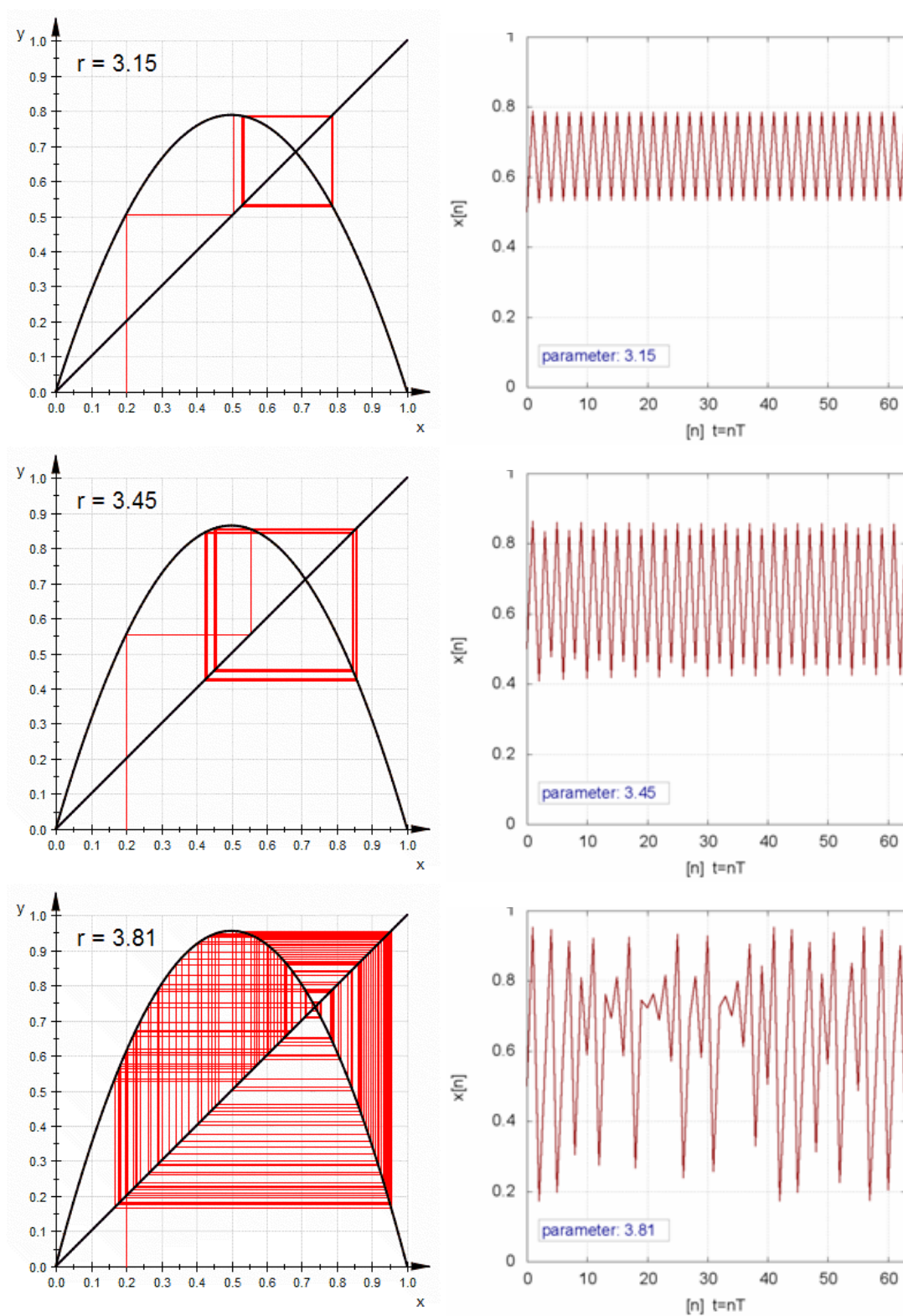


Figure 3.3: Behaviour of the logistic map for parameter values $\mu = 3.15$, $\mu = 3.45$, and $\mu = 3.81$. Figures from wikipedia.

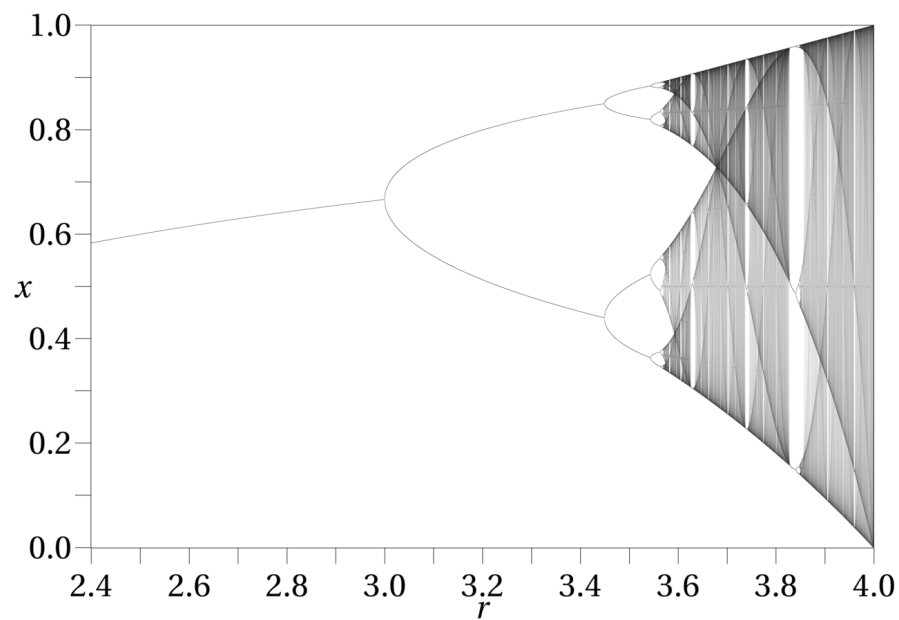


Figure 3.4: Bifurcation diagram for the logistic map. The attractor for any value of the parameter μ is shown on the vertical line at that μ . Figure from wikipedia.

4. Population dynamics: Stochastic models

The models in the previous chapters were all *deterministic* — a definite population $N(t_1)$ at time t_1 evolves, with certainty, into population $N(t_2)$ at later time $t_2 > t_1$. Alternatively, we might want to include stochasticity, or some “randomness” to our models. This might reflect:

- *Demographic stochasticity*, which describes the randomness that results from the inherently discrete nature of individuals. It has the largest impact on small populations.
- *Environmental stochasticity*, which describes the randomness resulting from any change that impacts an entire population (such as changes in the environment). Its impact does not diminish as populations become large.

Let $N(t)$ be number of individuals at time t . We consider this now be a stochastic variable with

$$p_n(t) = P[N(t) = n]; \quad n = 0, 1, 2, \dots \quad (4.1)$$

the probability that N takes the value n at time t . Our population models now become methods for finding the probabilities $p_n(t)$ and not $N(t)$.

4.1 Stochastic birth process

We begin by considering a population subject only to growth due to the birth of individuals. Let us assume that we know $p_n(t) \forall n$ at a given time t . Then consider system at later time $t + \Delta t$ with Δt small enough (we will soon take the limit $\Delta t \rightarrow 0$) that at most one creature can be born in the interval $t, t + \Delta t$. Connecting the probabilities at times $t + \Delta t$ and t , we can write

$$p_n(t + \Delta t) = P[1 \text{ birth} | N(t) = n - 1] p_{n-1}(t) + P[0 \text{ births} | N(t) = n] p_n(t) \quad (4.2)$$

To first order in Δt , let us expand the probability that a single individual gives birth in the interval Δt as $P[1 \text{ birth} | N(t) = 1] = \beta \Delta t + \mathcal{O}[(\Delta t)^2]$. From the binomial theorem, the probability that N individuals give rise to a single birth in the interval Δt is

$$\begin{aligned} P[1 \text{ birth} | N(t) = n] &= n[\beta \Delta t + \mathcal{O}[(\Delta t)^2]] [1 - [\beta \Delta t + \mathcal{O}[(\Delta t)^2]]]^{n-1} \\ &= n\beta \Delta t + \mathcal{O}[(\Delta t)^2]. \end{aligned} \quad (4.3)$$

The probability that there is no births is

$$P[0 \text{ birth} | N(t) = n] = [1 - [\beta \Delta t + \mathcal{O}[(\Delta t)^2]]]^n = 1 - n\beta \Delta t + \mathcal{O}[(\Delta t)^2]. \quad (4.4)$$

Thus, Eq. (4.2) becomes

$$p_n(t + \Delta t) = (n - 1)\beta\Delta t p_{n-1}(t) + [1 - n\beta\Delta t]p_n(t) + \mathcal{O}[(\Delta t)^2]. \quad (4.5)$$

Forming the differential quotient and taking the limit $\Delta t \rightarrow 0$ we obtain

$$\dot{p}_n = (n - 1)\beta p_{n-1} - n\beta p_n. \quad (4.6)$$

This holds true for all $n \geq 1$. For the special case of $n = 0$, we have

$$\dot{p}_0 = 0, \quad (4.7)$$

as there is no state “lower” than $n = 0$ to feed into $n = 0$ and a population of 0 can never give rise to new births. Typical initial conditions are to start with a definite population,

$$p_n = \delta_{n,n_0}. \quad (4.8)$$

Exercise 4.1 Show that the exact solution of Eq. (4.6) subject to Eq. (4.8) is given by

$$p_n(t) = \binom{n-1}{n_0-1} e^{-\beta n_0 t} (1 - e^{-\beta t})^{n-n_0}; \quad n \geq n_0 \quad (4.9)$$

and $p_n(t) = 0$ for $n < n_0$. ■

The system of ODEs specified by Eq. (4.6) is exactly solvable. For more complicated models of this type, this may not. Therefore, we now look at ways to extract useful information from equations like Eq. (4.6) without the full solution. The moments of the population distribution are given by

$$\langle N^k(t) \rangle \equiv \sum_{n=0}^{\infty} n^k p_n(t); \quad k = 1, 2, \dots \quad (4.10)$$

We can obtain information about the mean population ($k = 1$ in the above) by multiplying Eq. (4.6) with n , summing over all n , and noting that

$$\sum_{n=0}^{\infty} n \dot{p}_n(t) = \frac{d}{dt} \langle N(t) \rangle. \quad (4.11)$$

Doing this we obtain

$$\begin{aligned} \frac{d}{dt} \langle N(t) \rangle &= \beta \sum_{n=0}^{\infty} n(n-1) p_{n-1} - \beta \sum_{n=0}^{\infty} n^2 p_n \\ &= \beta \sum_{m=0}^{\infty} (m+1) m p_m - \beta \sum_{n=0}^{\infty} n^2 p_n \\ &= \beta [\langle N^2(t) \rangle - \langle N(t) \rangle] - \beta \langle N^2(t) \rangle \\ &= -\beta \langle N(t) \rangle. \end{aligned} \quad (4.12)$$

In the second line we have renamed $n - 1 = m$ and shifted the sum down by one. This is possible because the m factor in the summand means that the $m = 0$ term does not contribute. This stochastic birth model therefore gives a simple ODE for the mean $\langle N(t) \rangle$, which yields

$$\langle N(t) \rangle = e^{\beta t}. \quad (4.13)$$

The mean population in this model therefore shows exponential growth.

Exercise 4.2 Using Eq. (4.6) derive and solve an equation of motion for the variance of the population (its second cumulant)

$$\text{Var}[N(t)] \equiv \langle N^2(t) \rangle - \langle N(t) \rangle^2. \quad (4.14)$$

Compare your result with that obtainable from the probability distribution calculated directly in Ex. 4.1. ■

4.2 Stochastic birth and death processes

We now introduce death into our population model. Let $\mu\Delta t$ be the probability for a single individual to die in time Δt . Then, extending the above reasoning to both birth and death, we obtain

$$\begin{aligned} \dot{p}_0 &= \mu p_1 \\ \dot{p}_n &= +(n+1)\mu p_{n+1} + (n-1)\beta p_{n-1} - n(\beta + \mu)p_n; \quad n \geq 1. \end{aligned} \quad (4.15)$$

Exercise 4.3 Show that Eq. (4.15) conserves the norm of the probability distribution, ie. $\frac{d}{dt} \sum_{n=0} p_n(t) = 0$. ■

We are going to analyse this equation with the help of the *probability generating function*

$$F(z, t) \equiv \sum_{n=0} p_n(t) z^n. \quad (4.16)$$

The function $F(z, t)$ contains complete information about the probability distribution $p_n(t)$. It has several useful properties:

1. The probability of being extinct at time t is given by

$$p_0(t) = F(0, t) \quad (4.17)$$

2. Probabilities for higher n are obtained by taking derivatives and evaluating at the origin

$$p_n(t) = \frac{1}{n!} \left. \frac{\partial^n F}{\partial z^n} \right|_{z=0} \quad (4.18)$$

3. Derivatives of F evaluated at $z = 1$ can be used to calculate the moments (and cumulants) of the distribution $p_n(t)$. In particular

$$\langle N(t) \rangle = \left. \frac{\partial F}{\partial z} \right|_{z=1} \quad (4.19)$$

$$\text{Var}[N(t)] = \left[\frac{\partial^2 F}{\partial z^2} + \frac{\partial F}{\partial z} - \left(\frac{\partial F}{\partial z} \right)^2 \right]_{z=1} \quad (4.20)$$

We can derive an equation of motion for F by multiplying Eq. (4.15) with z^n and sum over all n . Identifying

$$\sum_{n=0} z^n \dot{p}_n(t) = \frac{\partial F}{\partial t} \quad (4.21)$$

$$\sum_{n=0} z^n n p_n(t) = \frac{\partial F}{\partial z} \quad (4.22)$$

we find that Eq. (4.15) becomes

$$\frac{\partial F}{\partial t} + (\beta z - \mu)(1 - z) \frac{\partial F}{\partial z} = 0, \quad (4.23)$$

which is a *partial differential equation* (PDE).

4.2.1 Method of characteristics

Eq. (4.23) is a first-order linear PDE. This type of equation we can solve using a (truncated version) of Lagrange's method of characteristics. Consider the PDE

$$a(z, t) \frac{\partial F}{\partial t} + b(z, t) \frac{\partial F}{\partial z} = 0 \quad (4.24)$$

The characteristic curves are defined through the solutions of the associated characteristic ODE

$$\frac{dz}{dt} = \frac{b(z, t)}{a(z, t)}. \quad (4.25)$$

Expressing the solutions of this equation in the form $k = v(z, t)$, with k a constant, the general solution to the PDE is

$$F(z, t) = W(v(z, t)), \quad (4.26)$$

with W an arbitrary function of its single argument.

In our case, the characteristic ODE reads

$$\begin{aligned} \frac{dz}{dt} &= (\beta z - \mu)(1 - z) \\ \Rightarrow (\beta - \mu)dt &= \left[\frac{\beta}{\beta z - \mu} + \frac{1}{1 - z} \right] dz. \end{aligned} \quad (4.27)$$

Integrating (and choosing the form of the constant for convenience)

$$\begin{aligned} (\beta - \mu)t &= \log(\beta z - \mu) - \log(1 - z) + \log k \\ \Rightarrow k &= e^{rt} \left(\frac{1 - z}{\beta z - \mu} \right), \end{aligned} \quad (4.28)$$

with $r = \beta - \mu$. Thus, general solution is

$$F(z, t) = W \left[e^{rt} \left(\frac{1 - z}{\beta z - \mu} \right) \right]. \quad (4.29)$$

With the initial probability of Eq. (4.1), the initial generating function is

$$F(z, t = 0) = z^{n_0}. \quad (4.30)$$

Equating this to the general solution at $t = 0$ gives the arbitrary function to be $W(v) = [(1 + \mu v)/(1 + \beta v)]^{n_0}$. Thus, the particular solution we require is

$$F(z, t) = \left[\frac{\mu(1 - z)e^{rt} + (\beta z - \mu)}{\beta(1 - z)e^{rt} + (\beta z - \mu)} \right]^{n_0}. \quad (4.31)$$

In deriving this result, we have assumed that $\mu \neq \beta$.

Exercise 4.4 Show that when $\mu = \beta$, the probability generating function reads

$$F(z, t) = \left[\frac{\beta t + (1 - \beta t)z}{(1 + \beta t) - \beta t z} \right]^{n_0}. \quad (4.32)$$

Our final result is therefore

$$F(z, t) = \begin{cases} \left[\frac{\mu(1-z)e^{rt} + (\beta z - \mu)}{\beta(1-z)e^{rt} + (\beta z - \mu)} \right]^{n_0}, & \mu \neq \beta \\ \left[\frac{\beta t + (1-\beta t)z}{(1+\beta t) - \beta t z} \right]^{n_0}, & \mu = \beta \end{cases}. \quad (4.33)$$

From Eq. (4.33) we can immediately extract the extinction probability

$$p_0(t) = F(0, t) = \begin{cases} \left[\frac{\mu(e^{rt} - 1)}{\beta e^{rt} - \mu} \right]^{n_0}, & \mu \neq \beta \\ \left[\frac{\beta t}{1 + \beta t} \right]^{n_0}, & \mu = \beta \end{cases}. \quad (4.34)$$

In the asymptotic limit, $t \rightarrow \infty$, for $\beta \leq \mu$, we find that $\lim_{t \rightarrow \infty} p_0(t) = 1$. In this case, our population faces certain doom because on average deaths exceed births. For $\beta > \mu$, we find $\lim_{t \rightarrow \infty} p_0(t) = (\mu/\beta)^{n_0}$. This means that, even with net growth, there is still a chance that the population goes extinct due to the possibility that a statistical fluctuation will come along that is large enough to wipe out the whole population. Small populations are particularly vulnerable in this respect.

From Eq. (4.19), we simply obtain the mean

$$\langle N(t) \rangle = n_0 e^{rt} \quad (4.35)$$

This grows exponentially with the rate r , similar to Sec. 2.1. We also obtain the variance

$$\text{Var}[N(t)] = \begin{cases} n_0 \frac{\beta + \mu}{\beta - \mu} e^{rt} (e^{rt} - 1), & \mu \neq \beta \\ 2n_0 \beta t, & \mu = \beta \end{cases}, \quad (4.36)$$

which, in the long-time limit, grows exponentially for $\mu \neq \beta$ and linearly for $\mu = \beta$.

4.3 Moment generating function

Calculating the higher-order moments from the probability generating function gets rather cumbersome. Use of the *moment generating function* makes this task easier. It is defined as

$$M(\chi, t) \equiv \sum_{n=0}^{\infty} e^{n\chi} p_n(t). \quad (4.37)$$

Then, the k th moment is then given simply as

$$\langle N^k(t) \rangle = \sum_{n=0}^{\infty} n^k p_n(t) = \left. \frac{\partial^k M}{\partial \chi^k} \right|_{\chi=0}. \quad (4.38)$$

This moment generating function can be simply obtained from the probability generating function as

$$M(\chi, t) = F(z = e^\chi, t). \quad (4.39)$$

We can also construct the *cumulant generating function* $G(\chi, t) = \log M(\chi, t)$, the k th derivative of which, evaluated at $\chi = 0$, gives the k th cumulant.

4.4 Outlook

The birth-death model we've studied in this chapter is of a type that is important in modelling in many branches of science. It is an example of a continuous-time Markov chain; in physics, you might here it referred to as a recombination-generation model. A more general “one-step process” model reads

$$\dot{p}_n = r_{n+1}p_{n+1} + g_{n-1}p_{n-1} - (r_n + g_n)p_n, \quad (4.40)$$

with general rates r_n and g_n for leaving state n . In our birth-death model, r_n and g_n were linear functions of n ; non-linear models also occur in ecology, see e.g.

- *Modelling biological populations in space and time*, E. Renshaw (Cambridge 1991).

5. Predator-prey interactions

The models we have discussed so far have all been single species models. Things get much more interesting when we consider multiple species, and the interaction between a predator species and its prey is one of the classic problems of mathematical ecology.

5.1 Lotka-Volterra predator-prey model

The Lotka-Volterra (LV) model (developed independently by Lotka (1925) and Volterra (1926)) is the simplest model of predator-prey interactions. Let P be density of the predators and N the density of prey species. The LV model then reads

$$\frac{dN}{dt} = rN - cNP = N(r - cP) \quad (5.1a)$$

$$\frac{dP}{dt} = -mP + bNP = P(-m + bN) \quad (5.1b)$$

with $r, m, c, b > 0$. The first term in each of these equations means that, without interaction ($c = b = 0$) the prey population will grow exponentially with rate r , and the predator will decline exponentially with rate m . The two populations are coupled by a trophic interaction ($c \neq 0$ and $b \neq 0$) such that the presence of predators ($P > 0$) leads to a decrease in the growth rate of the prey, and conversely that the presence of prey $N > 0$ has a positive effect on the growth rate of the predator. The opposition of these two effects means that this is an *antagonistic* interaction, where one species gains at the others expense.

We start our analysis by rescaling. Setting

$$x = \frac{b}{m}N; \quad y = \frac{c}{r}P; \quad \tau = rt; \quad \alpha = \frac{m}{r}, \quad (5.2)$$

gives

$$\frac{dx}{d\tau} = f(x, y) = x(1 - y); \quad (5.3a)$$

$$\frac{dy}{d\tau} = g(x, y) = \alpha y(x - 1), \quad (5.3b)$$

which now has but a single parameter α .

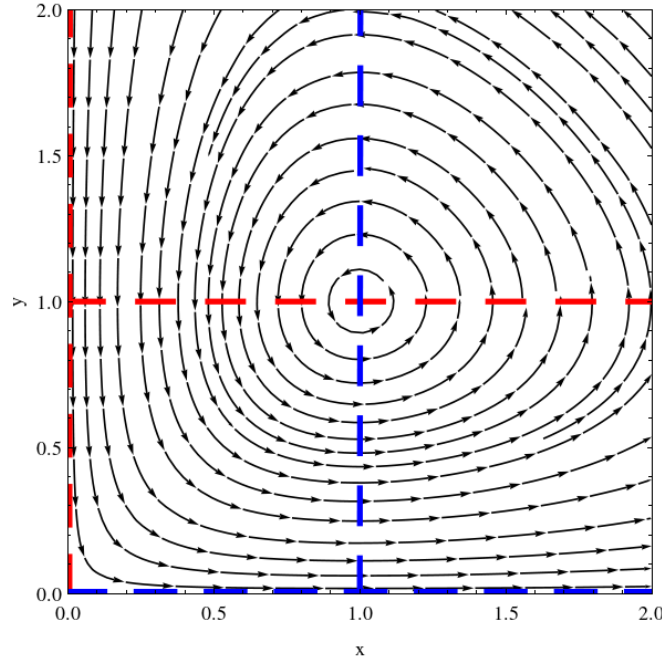


Figure 5.1: Sketch of vector field representation of Lotka-Volterra model ($\alpha = 1$). Dashed lines are nullclines (red: $dx/d\tau = 0$; blue: $dy/d\tau = 0$). Arrows represent direction of phase-space flow.

5.1.1 Phase-space analysis

Qualitative understanding of the behaviour of models of this type can be obtained from a phase-space analysis in which we plot prey population as the x -axis and predator population as the y -axis, see Fig. 5.1. To analyse Eq. (5.3), the first thing to plot are the *nullclines*. These are the lines along which the derivatives vanish, i.e. for which $\frac{dx}{d\tau} = 0$ and $\frac{dy}{d\tau} = 0$. For the LV model, setting $\frac{dx}{d\tau} = 0$ gives the nullclines $x = 0$ and $y = 1$; setting $\frac{dy}{d\tau} = 0$ gives the nullclines $y = 0$ and $x = 1$. Where the nullclines from different derivatives cross, we obtain a fixed point of the model. The $x = 1$ and $y = 1$ nullclines in Fig. 5.1 cross at positive fixed point $(x_*, y_*) = (1, 1)$. The nullclines also cross at the point $(x, y) = (0, 0)$, where both species are extinct. Further detail is added to the phase space plot by sketching the direction of motion at select points in the space,, i.e. we are plot the vector field (f, g) . For the LV model, the direction of the arrows is given by

$$\frac{dy}{dx} = \frac{dy}{d\tau} \frac{d\tau}{dx} = \frac{\alpha y(x-1)}{x(1-y)}. \quad (5.4)$$

If doing this by hand, an easy place to start is along the nullclines, as the vector field is necessarily perpendicular to the respective axis. From the pattern of the arrows in Fig. 5.1, we see that the generic behaviour of the LV model appears to be oscillations of the two populations about the positive fixed point.

5.2 Stability analysis of planar systems

To understand fully the behaviour of systems such as the LV model, we need to consider linear stability in two dimensions. The following results generalise to more dimensions/species. Consider an autonomous model for two interacting populations x and y of the general form

$$\frac{dx}{d\tau} = f(x, y); \quad \text{and} \quad \frac{dy}{d\tau} = g(x, y) \quad (5.5)$$

where $f : \mathbb{R}^{\geq 0} \times \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}$ and $g : \mathbb{R}^{\geq 0} \times \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}$ are continuous and continuously differentiable. With (x_*, y_*) a fixed point of Eq. (5.5) such that

$$f(x_*, y_*) = g(x_*, y_*) = 0, \quad (5.6)$$

we investigate stability by setting

$$x(t) = x_* + u(t); \quad y(t) = y_* + v(t), \quad (5.7)$$

and assuming u and v are sufficiently small that we may expand Eq. (5.5) to first order in them. We have

$$\frac{du}{d\tau} = \frac{dx}{d\tau} \approx u(t)\partial_x f|_* + v(t)\partial_y f|_*, \quad (5.8)$$

where the notation $|_*$ means evaluation at $x = x_*$ and $y = y_*$, and where we have used $f(x_*, y_*) = 0$. A similar result pertains for $dv/d\tau$. Writing them together in matrix form, we have

$$\frac{d}{dt} \begin{pmatrix} u \\ v \end{pmatrix} = A \begin{pmatrix} u \\ v \end{pmatrix} \quad (5.9)$$

where matrix

$$A = \begin{pmatrix} \partial_x f & \partial_y f \\ \partial_x g & \partial_y g \end{pmatrix}_*, \quad (5.10)$$

is the Jacobian matrix. In an ecological context it is known as the *community matrix*. The stability of the fixed point depends on the eigenvalues of A . These are obtained by solving the secular equation

$$\det(A - \lambda I) = 0. \quad (5.11)$$

For a fixed point to be stable, all solutions of this equation must have a negative real part. In the two-by-two case, the character of the fixed points can be characterised by the trace $\mathcal{T} = \text{Tr} A$ and determinant $\Delta = \det A$. In these terms, the secular equation reads

$$\lambda^2 - \mathcal{T}\lambda + \Delta = 0. \quad (5.12)$$

and the two eigenvalues are thus

$$\lambda_{\pm} = \frac{\mathcal{T} \pm \sqrt{\mathcal{T}^2 - 4\Delta}}{2}. \quad (5.13)$$

If the real parts of $\lambda_{\pm}$ are both negative, the fixed point is an “attractor” and stable; if one real part is negative and the other positive, the fixed point is a saddle point (unstable); if both real parts are positive, the fixed point is a “repellor” and unstable. Depending on whether the eigenvalues have imaginary parts or not, a repellor or attractor may either be a focus (finite imaginary parts) or a node (purely real eigenvalues). The behaviours associated with these various forms of fixed points are sketched in Fig. 5.2. From this figure, we see that there are shortcuts for the stability of a 2D system

- If $\Delta < 0$, then eigenvalues are real, nonzero and have opposite sign, and the fixed point is a saddle point and unstable;
- If $\Delta > 0$ and $\mathcal{T} < 0$, then eigenvalues are either a pair of negative real numbers or a complex conjugate pair with negative real part. Either way, the fixed point is stable;
- If $\Delta > 0$ and $\mathcal{T} > 0$, then eigenvalues are either a pair of positive real numbers or a complex conjugate pair with positive real part. Either way, the fixed point is unstable.

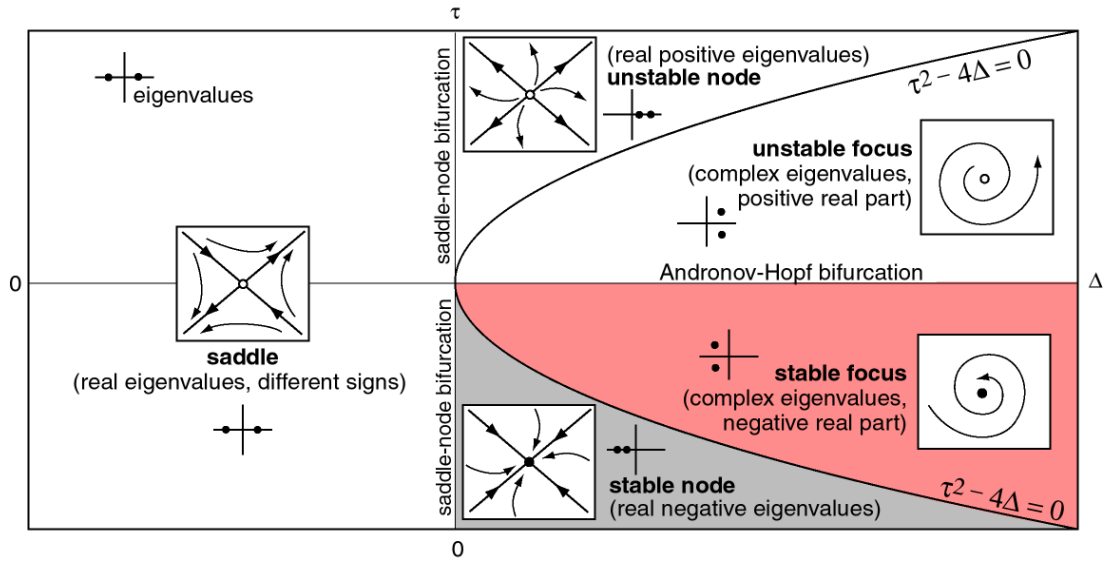


Figure 5.2: Characterisation of fixed points for a planar system in terms of the trace, (here written as τ , same as $\mathcal{T}$ in the text!), and determinant, Δ , of the community matrix. Figure from scholarpedia.

- If discriminant $\tau^2 - 4\Delta < 0$, the eigenvalues will have nonzero imaginary parts.

Note that the eigenvectors of the community matrix give the directions of the stable and unstable motion near the fixed point.

One important edge case in Fig. 5.2 exists along line $\tau = 0$ with $\Delta > 0$, as here the eigenvalues are purely imaginary. In this case, the fixed point is referred to as a *centre*, and trajectories sufficiently close to a centre neither collapse towards or diverge away from it, but rather just circulate around it in closed orbits.

5.3 Stability analysis of Lotka-Volterra model

We now return to the (rescaled) Lotka-Volterra model of Eq. (5.3). We can immediately find the Jacobian matrix

$$A = \begin{pmatrix} (1-y) & -x \\ \alpha y & \alpha(x-1) \end{pmatrix}. \quad (5.14)$$

Evaluating this at the two fixed points of the models gives

(x_*, y_*)	community matrix A	Eigenvalues	Stability type
$(0, 0)$	$\begin{pmatrix} 1 & 0 \\ 0 & -\alpha \end{pmatrix}$	$\lambda = 1, -\alpha$	Saddle
$(1, 1)$	$\begin{pmatrix} 0 & -1 \\ \alpha & 0 \end{pmatrix}$	$\lambda = \pm i\sqrt{\alpha}$	Centre

The saddle point at the origin means that predator-only populations will die out, but that prey-only populations will increase without bound. The positive fixed point has purely imaginary eigenvalues, and thus is a *centre*. Fig. 5.3a shows closed trajectories encircling it, corresponding to periodic oscillations of the populations as shown in Fig. 5.3b. Another pertinent feature of this model is that the peaks of the predator's oscillation lag slightly behind those of the prey's oscillation.

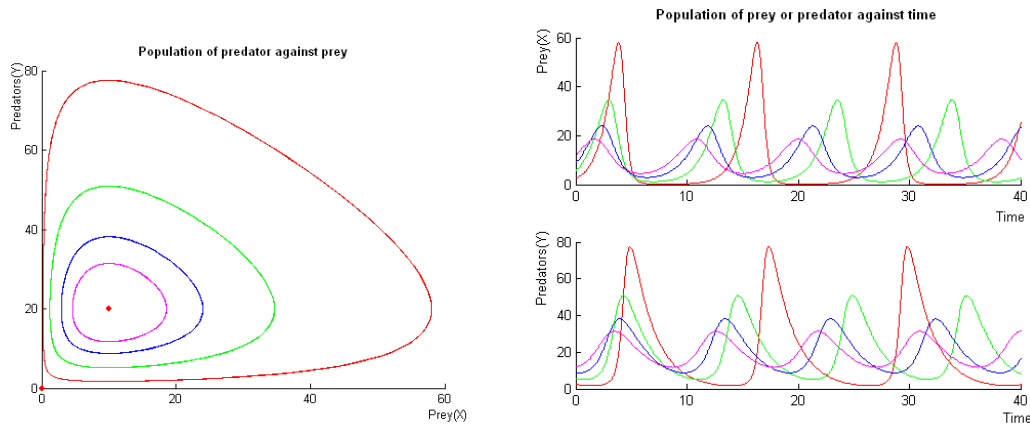


Figure 5.3: (a) Phase space trajectories and (b) time series of prey and predator populations according to LV model for typical parameters. Different colours correspond to different initial conditions. Figures from openwetware.org.

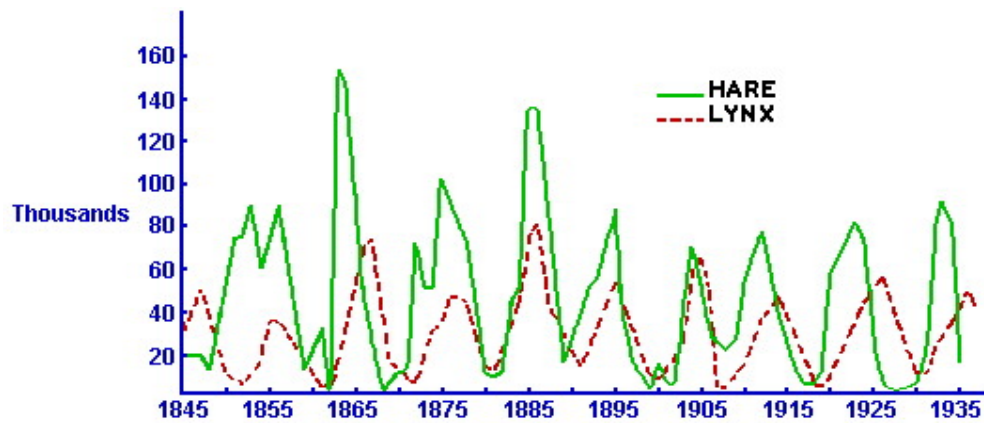


Figure 5.4: Oscillations in Canadian Lynx and Snowshoe Hare populations. Plot of data from classic paper: “The ten-year cycle in numbers of lynx in Canada.” by Elton and Nicholson J. Anim. Ecol. **11** 215 (1942). Figure from duke.edu.

Fig. 5.4 shows data for what is taken to be a classic textbook example of predator-prey dynamics — long-term populations of Canadian lynx and snowshoe hare. On the face of it, this data bears out many features of the LV model — there are clear oscillations of the two species, and also some lags between hare and lynx peaks. More careful analysis this data, however, means the link with LV dynamics is not so clear, see

- M.E. Gilpin, . *Do Hares Eat Lynx?* The American Naturalist, **107** 727 (1973).

Aside from the question of how well this model describes any given set of observations, there is also a theoretical objection to the LV model. The presence of a centre, and the neutral stability that this represents, means that the model is structurally unstable — small changes to its structure can lead to dramatic changes in the the behaviour.

5.4 Predator-prey with logistic growth of prey

Many improvements to the LV model are possible. In this section, we consider that, rather than increasing without bound, the prey are subject to a carrying capacity and hence logistic growth. Our modified LV model reads

$$\frac{dN}{dt} = rN(1 - N/K) - cNP \quad (5.15a)$$

$$\frac{dP}{dt} = -mP + bNP \quad (5.15b)$$

where K is the carrying capacity for the prey. We rescale to obtain

$$\frac{dx}{d\tau} = x(1 - x - y) \quad (5.16a)$$

$$\frac{dy}{d\tau} = \beta(x - \alpha)y \quad (5.16b)$$

in terms of the two positive parameters α and β .

Exercise 5.1 Find the rescaling that leads from Eq. (5.15) to Eq. (5.16). ■

Exercise 5.2 Show that the fixed points of Eq. (5.16) are

$$(x_*, y_*) = (0, 0); \quad (x_*, y_*) = (1, 0); \quad (x_*, y_*) = (\alpha, 1 - \alpha). \quad (5.17)$$

Then plot the nullclines and fixed points of this model in the xy plane. ■

To investigate stability, we calculate the Jacobian

$$A = \begin{pmatrix} 1 - 2x - y & -x \\ \beta y & \beta(x - \alpha) \end{pmatrix}, \quad (5.18)$$

and obtain the following fixed-point properties:

(x_*, y_*)	community matrix A	Eigenvalues	Stability type
$(0, 0)$	$\begin{pmatrix} 1 & 0 \\ 0 & -\beta\alpha \end{pmatrix}$	$\lambda = 1, -\beta\alpha$	Saddle
$(1, 0)$	$\begin{pmatrix} -1 & -1 \\ 0 & \beta(1 - \alpha) \end{pmatrix}$	$-1, \beta(1 - \alpha)$	saddle or stable node
$(\alpha, 1 - \alpha)$	$\begin{pmatrix} -\alpha & -\alpha \\ \beta(1 - \alpha) & 0 \end{pmatrix}$	$\lambda = \frac{-\alpha \pm \sqrt{\alpha^2 + 4\alpha\beta(\alpha - 1)}}{2}$	see text

This analysis shows that behaviour near the origin is the same as for the LV model. The stability of the two remaining fixed points depends on the parameters. For $\alpha < 1$, the fixed point $(1, 0)$ is unstable and that at $(\alpha, 1 - \alpha)$ is stable. For $\alpha > 1$, the stability reverses. When stable, fixed point $(1, 0)$ is a node; when unstable it is a saddle point. Furthermore, by looking at the discriminant, we see that for $\alpha > 4\beta/(1 + 4\beta)$ the fixed point $(\alpha, 1 - \alpha)$ is a node; otherwise it is a spiral focus. Phase space plots for this model in these two regimes are given in Fig. 5.5.

Thus, the behaviour here is very different to that of the LV model — in particular the oscillations are gone. In their stead we obtain a transition, as α increases we move from a stationary state where both predator and prey are present ($\alpha < 1$) to a state stationary state where predator is extinct ($\alpha > 1$).

It therefore appears that the oscillations the LV model have been vanquished by the inclusion of a carrying capacity, no matter its size — this vindicates our concerns about the LV

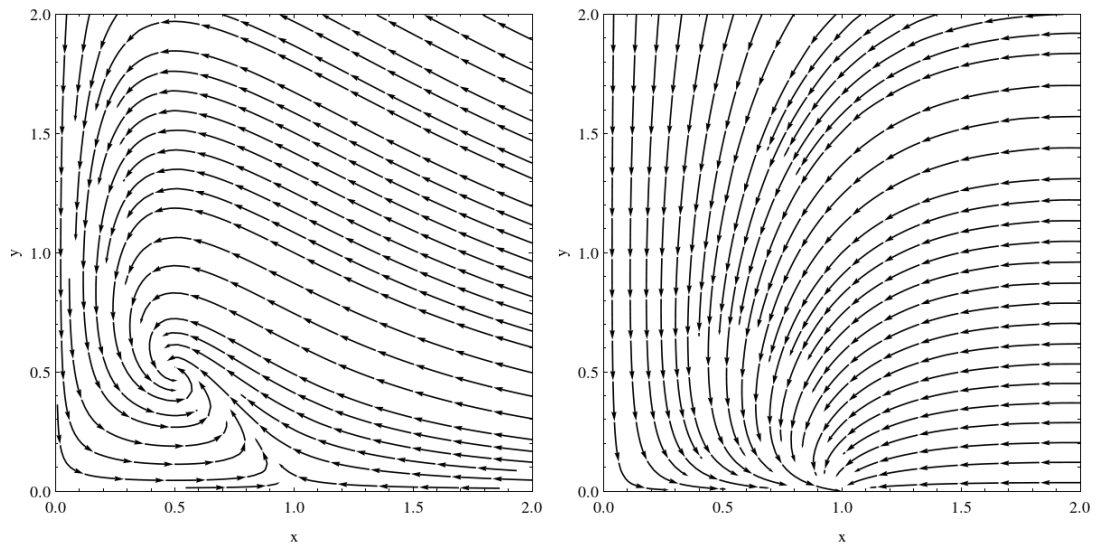


Figure 5.5: Sketch of vector field representation of Lotka-Volterra model with logistic growth with $\beta = 1$. LEFT: $\alpha = 1/2$. Non-trivial fixed point is a stable focus. RIGHT: $\alpha = 2$. Only stable fixed point in the $(x, y) = (1, 0)$ with predator species exist.

model's structural instability. We can prove that there are no oscillations in this model with the help of the following theorem.

Theorem 5.4.1 *Bendixson-Dulac negative criterion.* Consider the autonomous dynamical system

$$\frac{dx}{d\tau} = f(x, y); \quad \frac{dy}{d\tau} = g(x, y) \quad (5.19)$$

with f and g continuously-differentiable functions on some simply-connected domain of $D \in \mathbb{R}^2$. Let B be any smooth function on D (the *Dulac function*). Then, if

$$\nabla \cdot (Bf, Bg) = \frac{\partial(Bf)}{\partial x} + \frac{\partial(Bg)}{\partial y} \quad (5.20)$$

is of one sign ($\neq 0$) in D , then there cannot be a closed orbit contained within D .

Proof. See Kot p124-135. ■

Exercise 5.3 Use the Bendixson-Dulac theorem with Dulac function $B = 1/(xy)$ to show that the modified LV model of Eq. (5.16) has no periodic oscillations. ■

5.5 Functional Response models

Another way in which we might look to improve the LV model is to change the interaction term to reflect more realistically how the feeding rate of the predator depends on the prey density. Let P be the predator density and N that of the prey. We can write the interaction term that occurs in equation of motion for the predators as $+Pf(N)$ which, as before, grows linearly with predator density, but now includes the function $f(N)$ to represent the food intake rate. Following the work of C. S. Holling (1959) the three main types of functional response traditionally considered are:

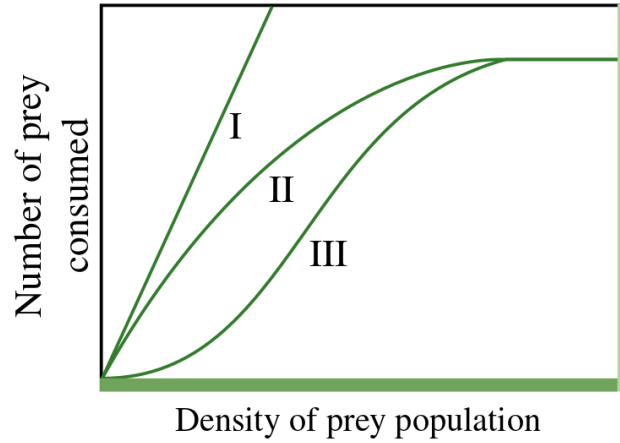


Figure 5.6: Sketch of vector field representation of Lotka-Volterra. Dashed lines are nullclines. Arrows represent direction of phase-space flow. Figure from wikipedia.

- *Type I*: Linear response, as in original LV model. The feeding rate is

$$f(N) = aN, \quad (5.21)$$

with a a constant.

- *Type II*: A hyperbolic function that saturates due to the time it takes for predators to handle their prey. The response can be written

$$f(N) = \frac{aN}{1 + ahN}, \quad (5.22)$$

where a is the “attack rate” and h the “handling time”

- *Type III*: is a sigmoid curve that saturates and also reflects that prey are particularly hard to obtain at low densities. A typical form is

$$f(N) = \frac{cN^2}{a^2 + N^2}, \quad (5.23)$$

with parameters a and c .

The functions are compared in Fig. 5.6. Since Holling, other functional responses have been proposed, but these are the main three.

5.6 Rosenzweig-McArthur model

Rosenzweig and McArthur (1963) considered a predator-prey model where the interaction is via a Hollings type-II response. The model reads

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) - \frac{cNP}{a + N} \quad (5.24a)$$

$$\frac{dP}{dt} = -mP + \frac{bNP}{a + N}. \quad (5.24b)$$

(NB: The prey exhibits logistic growth and the predators suffer exponential decline). This model can be rescaled to give

$$\frac{dx}{d\tau} = x \left(1 - \frac{x}{\gamma} \right) - \frac{xy}{1 + x} \quad (5.25a)$$

$$\frac{dy}{d\tau} = \beta \left(\frac{x}{1 + x} - \alpha \right) y. \quad (5.25b)$$

5.6.1 Analysis and Hopf bifurcation

Not examined -- see Murray 88-94, Kot pages 127-139

6. More species interactions

Whilst the predator-prey is perhaps the most obvious interaction two-species interaction, nature abounds with other forms of interaction. These we may classify into three groups

- *Antagonistic*: An interaction where one species benefits to detriment of the other. Predation and parasitism as of this type.
- *Competitive*: An interaction that is detrimental to both species, e.g. competition for a shared resource.
- *Cooperative/Mutualistic*: An interaction that benefits both species, e.g. pollination, cleaner fish.

This information can be summarised in the following table

interaction type	species 1	species 2
antagonistic	+	–
competition	–	–
mutualistic	+	+

where the signs denote whether an increase in population of the other species has a positive or negative effect on the species in question.

In this chapter we look at two-species models of competition and mutualism based of the Lotka-Volterra foundation of the previous chapter. In the final section, we move beyond two species and take a look at ecological networks.

6.1 Competition

Consider two species with populations $N_i; i = 1, 2$ that we assume compete with one another. Taking inspiration from the Lotka-Volterra predator-prey model, an obvious starting point is the following model

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1 + a_1 N_2}{K_1} \right) \quad (6.1a)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2 + a_2 N_1}{K_2} \right) \quad (6.1b)$$

where $r_i > 0$ and $K_i > 0$ are the growth rates and carrying capacities of the two species. The a_i are positive constants that reflect the effects of competition on species i (this parameterisation is chosen such that $a_i < 1$ denotes a situation where the effects of interspecific competition is weaker than intraspecific competition, and for $a_i > 1$ vice versa).

Exercise 6.1 Find the rescaling that puts Eq. (6.1) in the form

$$\frac{dx}{d\tau} = f(x, y) = x(1 - x - \alpha y) \quad (6.2a)$$

$$\frac{dy}{d\tau} = g(x, y) = \rho y(1 - y - \beta x) \quad (6.2b)$$

with positive parameters $\alpha, \beta, \rho > 0$. ■

Note that the signs of the interaction here, negative in both equations, is commensurate with the deleterious effect of competition for both species. Analysis of this model starts with the nullclines

$$\frac{dx}{d\tau} = 0 : \Rightarrow x = 0 \text{ or } (1 - x - \alpha y) = 0 \quad (6.3)$$

$$\frac{dy}{d\tau} = 0 : \Rightarrow y = 0 \text{ or } (1 - y - \beta x) = 0, \quad (6.4)$$

and the community matrix

$$A = \begin{pmatrix} 1 - 2x - \alpha y & -\alpha x \\ -\rho\beta y & \rho(1 - 2y - \beta x) \end{pmatrix}. \quad (6.5)$$

From these we find the following fixed points and their stabilities

(x_*, y_*)	community matrix A	Eigenvalues	Stability type
$(0, 0)$	$\begin{pmatrix} 1 & 0 \\ 0 & \rho \end{pmatrix}$	$\lambda = 1, \rho$	unstable node
$(1, 0)$	$\begin{pmatrix} -1 & -\alpha \\ 0 & \rho(1 - \beta) \end{pmatrix}$	$\lambda = -1, \rho(1 - \beta)$	saddle ($\beta < 1$) or stable node ($\beta > 1$)
$(0, 1)$	$\begin{pmatrix} 1 - \alpha & 0 \\ -\rho\beta & -\rho \end{pmatrix}$	$\lambda = -\rho, 1 - \alpha$	saddle ($\alpha < 1$) or stable node ($\alpha > 1$)
$\left(\frac{1-\alpha}{1-\alpha\beta}, \frac{1-\beta}{1-\alpha\beta}\right)$	$\begin{pmatrix} -x_* & -\alpha x_* \\ -\rho\beta y_* & -\rho y_* \end{pmatrix}$	see text	see text — f.p. can be infeasible

In discussing these results we may, without loss of generality, assume $\alpha \leq \beta$ (can always swap roles of x and y to make this true). There are then three distinct parameter regimes:

- *Weak competition:* $\alpha \leq \beta < 1$. The fixed point $(x_*, y_*) = \left(\frac{1-\alpha}{1-\alpha\beta}, \frac{1-\beta}{1-\alpha\beta}\right)$ is feasible. Furthermore, the trace and determinant of the community matrix are $\mathcal{T} = -x_* - \rho y_*$ and $\Delta = \rho x_* y_* (1 - \alpha\beta)$. Since $x_*, y_*, (1 - \alpha\beta) > 0$, these have the signs $\mathcal{T} < 0$ and $\Delta > 0$ and thus this fixed point is stable. The other three fixed points are unstable. Thus, in this case, coexistence of the two species is possible. This makes some intuitive sense, as α and β are “small” here and interaction between the species is weak.
- *Competitive exclusion:* $\alpha < 1 < \beta$. The fixed point where the species coexists contains at least one negative population, and is thus infeasible. The only feasible, stable point in this case is $(1, 0)$. Thus, in this case, the competitive advantage of species 1 over species 2 means that almost all initial conditions lead to the extinction of species 2.

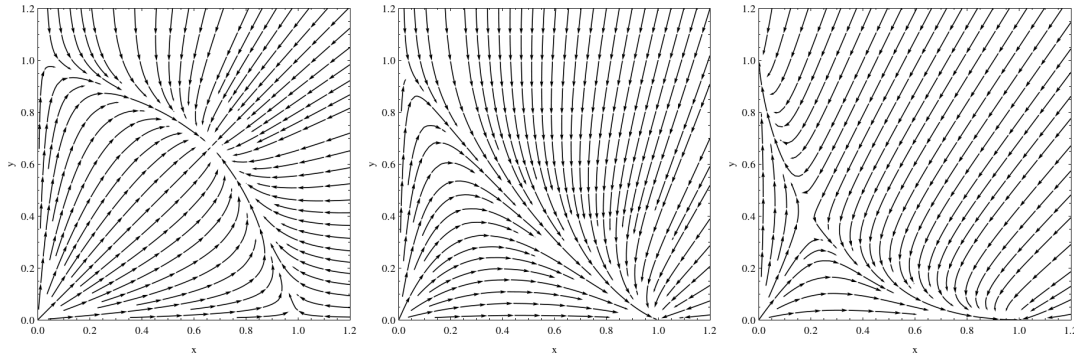


Figure 6.1: Sketch of vector field representation of LV competition model with $\rho = 1$. LEFT: $\alpha = \beta = 1/2$; weak competition. MIDDLE: $\alpha = 1/2$, $\beta = 2$ competitive exclusion. RIGHT: $\alpha = 2$, $\beta = 3$; strong competition.

- *Strong competition:* $1 \leq \alpha \leq \beta$. Here, coexistence becomes once again possible as fixed point $(x_*, y_*) = \left(\frac{1-\alpha}{1-\alpha\beta}, \frac{1-\beta}{1-\alpha\beta}\right)$ is feasible once again. However, since $(1 - \alpha\beta) < 0$, this fixed point is unstable. In contrast, the two fixed points at $(1, 0)$ and $(0, 1)$ are both stable nodes, and both represent possible long-term states of the system,. Which one the system eventually ends up in depends on the initial conditions (basin of attraction).

Typical phase space plots for each of these scenarios is given in Fig. 6.1.

6.2 Mutualism

Form wikipedia:

Mutualism is the way two organisms of different species exist in a relationship in which each individual benefits from the activity of the other.

The cleaner fish and wrasse pictured in the chapter image is one example of mutualism, but there are countless others. One biologically important classification of mutualistic relationships is as either obligate, where a species needs its mutualistic partner to survive,; or facultative, where presence of the partner species provides some non-essential benefit.

We consider first facultative mutualism and again we take a modification of the LV equations as an opening approach. We employ the rescaled form of Eq. (6.2) but now with positive, rather than negative, signs for the interactions. Our model of facultative mutualism reads

$$\frac{dx}{d\tau} = x(1 - x + \alpha y); \quad (6.6a)$$

$$\frac{dy}{d\tau} = \rho y(1 - y + \beta x), \quad (6.6b)$$

with positive parameters $\alpha, \beta, \rho > 0$. Fixed-point analysis yields:

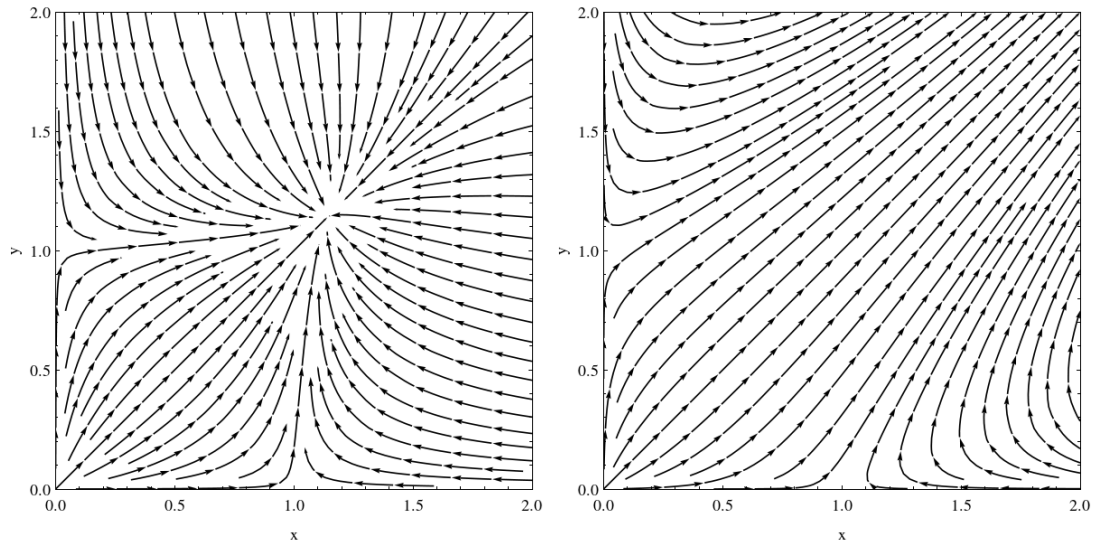


Figure 6.2: Sketch of vector field representation of facilitative mutualism with $\rho = 1$. LEFT: $\alpha = \beta = 1/8$. Stable node at $(8/7, 8/7)$. RIGHT: $\alpha = \beta = 2$.

(x_*, y_*)	community matrix A	Eigenvalues	Stability type
$(0, 0)$	$\begin{pmatrix} 1 & 0 \\ 0 & \rho \end{pmatrix}$	$\lambda = 1, \rho$	unstable node
$(1, 0)$	$\begin{pmatrix} -1 & \alpha \\ 0 & \rho(1 + \beta) \end{pmatrix}$	$\lambda = -1, \rho(1 + \beta)$	saddle
$(0, 1)$	$\begin{pmatrix} 1 + \alpha & 0 \\ \rho\beta & -\rho \end{pmatrix}$	$\lambda = -\rho, 1 + \alpha$	saddle
$\left(\frac{1+\alpha}{1-\alpha\beta}, \frac{1+\beta}{1-\alpha\beta}\right)$	$\begin{pmatrix} -x_* & \alpha x_* \\ \rho\beta y_* & -\rho y_* \end{pmatrix}$	see text	see text — can be infeasible

and reveals two regimes:

- *Stable mutualism:* $\alpha\beta < 1$. The trace and determinant of the community matrix for the fixed point $\left(\frac{1+\alpha}{1-\alpha\beta}, \frac{1+\beta}{1-\alpha\beta}\right)$ take the same form as in the competitive model, $\mathcal{T} = -x_* - \rho y_*$ and $\Delta = \rho x_* y_* (1 - \alpha\beta)$, and here have the signs $\mathcal{T} < 0$ and $\Delta > 0$. Thus, this fixed point is the only stable one.
- *Instability:* $\alpha\beta > 1$. The fixed point $\left(\frac{1+\alpha}{1-\alpha\beta}, \frac{1+\beta}{1-\alpha\beta}\right)$ becomes unstable. There is no nontrivial fixed point in the first quadrant and, for almost all initial conditions, the populations undergo unlimited growth in what May (1981) describes as ‘an orgy of mutual benefaction’.

Phase space plots of these behaviours can be seen in Fig. 6.2.

In the homework, you will study a model for obligate mutualism based on the Lotka-Volterra-formalism. As in the facultative case, final outcomes include ever-increasing populations. Thus, unlike in the competitive case, where LV-type models generally give sensible results, for mutualism, the equivalent models give unrealistic results. This arises from the fact that the benefits of mutualism are unbounded in these models. This defect can be rectified with other models of

cooperation. e.g. model of Wolin and Lawlor (1984):

$$\begin{aligned}\frac{dN_1}{dt} &= N_1 \left(r_1 - d_1 N_1 - \frac{b_1 N_1}{1 + \alpha_{12} N_2} \right) \\ \frac{dN_2}{dt} &= N_2 \left(r_2 - d_2 N_1 - \frac{b_2 N_2}{1 + \alpha_{21} N_1} \right).\end{aligned}\tag{6.7}$$

In this model, increases in the population of the mutualistic partner lead to an effective increase in the carrying capacity. This model gives a stable fixed point with finite populations.

6.3 Ecological networks

Not examined -- for your own edification, you should read:

- R. May, 1972 *Will a Large Complex System be Stable?*
- R. May, 1973 *Qualitative Stability in Model Ecosystems*

